ShopSense: Hybrid Semantic Product Discovery Engine with Association-Rule Recommendations

DBMS Experiential Learning, Level 3 Project Report Database option 2: **SQL + Vector database** (PostgreSQL 17 with pgvector) · Web interface: **Next.js**

Team	Member A (transactions, integrity, security & analytics) · Member B (search, vectors & recommendations)
Stack	PostgreSQL 17.11 · pgvector 0.8.7 · Next.js 15 (App Router) · TypeScript · transformers.js (all-MiniLM-L6-v2)
Repository	*public GitHub link to be added by the team*

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1. Objectives

Online search is usually *lexical*: it returns products whose text contains the typed words. It fails when shoppers describe a **need** rather than a product name ("something to keep my dog's teeth clean"), and pure semantic search fails in the opposite direction on exact identifiers. The project builds a marketplace whose database layer solves both:

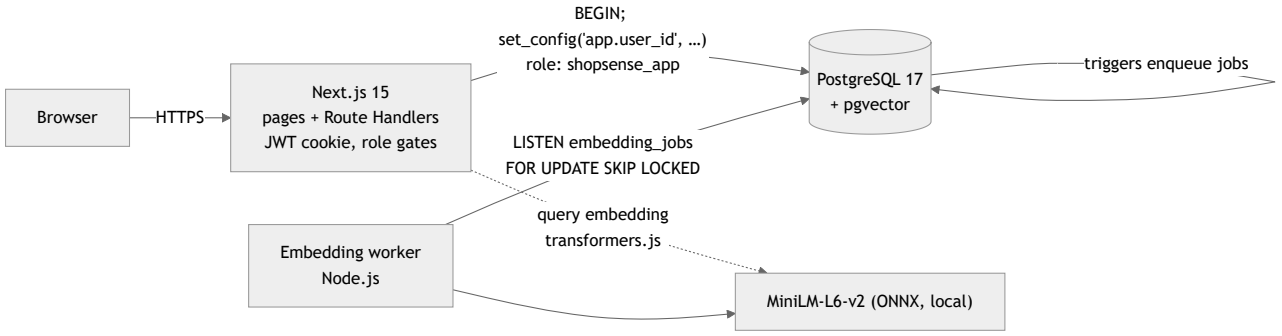
1. **Hybrid retrieval** that fuses PostgreSQL full-text ranking with vector similarity over product descriptions and customer reviews, in a single SQL function, with relational filters (price, category, rating, stock).
2. **Measured search quality**: lexical, semantic and hybrid retrieval compared on labelled queries, including a held-out set, rather than claimed.
3. **Database-enforced integrity and security**: constraints and domains, triggers, atomic and concurrency-safe transactions, role-based access enforced by Row-Level Security.
4. **Analytics** for sellers and administrators: revenue trends, association rules ("frequently bought together"), inventory alerts.

2. Functional requirements

ID	Requirement	Role
F1	Register / sign in; passwords hashed in the database; sessions as signed cookies	all
F2	Search products in natural language; modes keyword / semantic / hybrid; filters (category, price, rating, stock); explanation of why a product matched; side-by-side comparison of the three modes	all
F3	Typo-tolerant title suggestions	all
F4	Product page: details, rating distribution, reviews (verified-purchase badge), similar products, frequently-bought-together	all
F5	Cart, atomic checkout with address and simulated payment, order history, cancellation with stock restoration	customer
F6	Write a review (verified status computed by the database)	customer
F7	Personalised recommendations from purchase history	customer
F8	Manage own products: create, edit, price, stock, thresholds; restock; see embedding status	seller
F9	Seller dashboard: revenue trend, month-over-month growth, moving averages, top products with ranks, low-stock alerts, orders to fulfil with status progression	seller
F10	Platform dashboard: revenue, category roll-up, seller leaderboard, search analytics, zero-result queries, embedding queue	admin
F11	User management (activate/deactivate)	admin
F12	Search-quality benchmark results; concurrency lab; `EXPLAIN ANALYZE` viewer	admin

Non-functional: no credentials in source (environment variables), input validation at the database level, least privilege database role, reproducible seed data.

3. System architecture



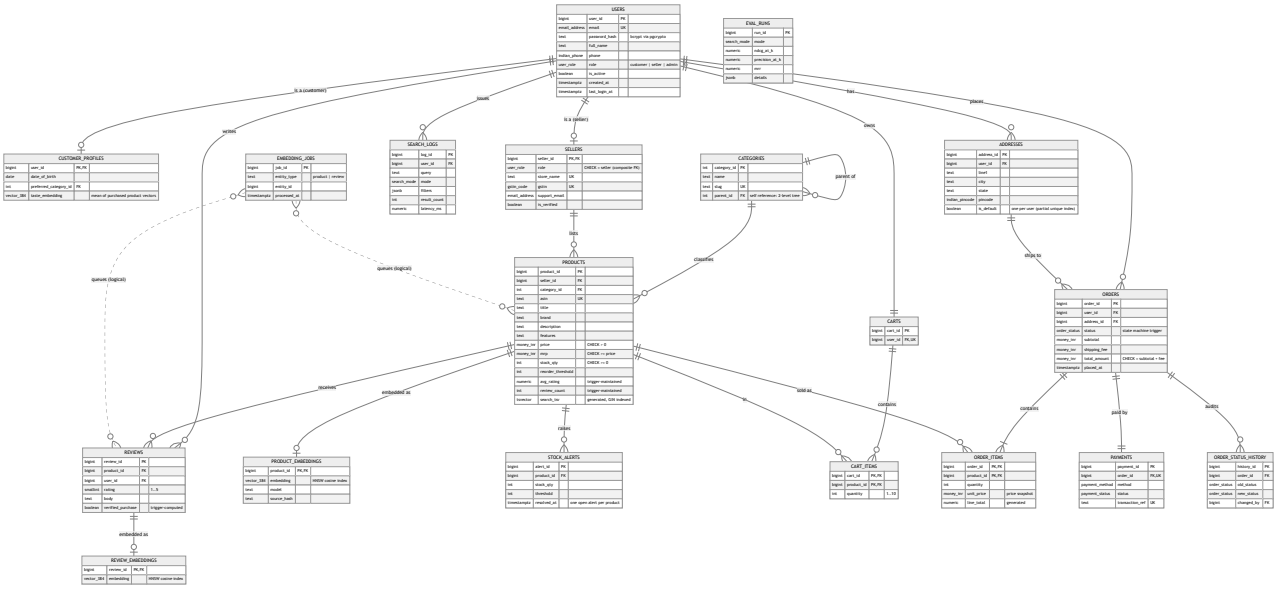
- **Next.js** serves pages (server components query the database directly through a data layer) and JSON route handlers. Every database interaction runs inside a transaction that first sets `app.user_id`. Row-Level Security policies use it, so the *same SQL* returns different rows for different users.
- **Role separation.** The web app connects as `shopsense_app` (no superuser, no `BYPASSRLS`, column-level grants; it cannot read `password_hash`). Setup scripts and the embedding worker use a separate owner role.
- **Embedding worker** drains a trigger-filled queue, so a seller editing a product description makes it semantically searchable again within seconds, without any application code remembering to do so.

4. Database choice and justification

Data	Technology	Justification
Users, sellers, catalog, prices, stock, carts, orders, payments, audit	**PostgreSQL (relational)**	Structured, strongly related data; needs ACID transactions (checkout must never oversell), foreign keys, constraints, joins and aggregate analytics.
Product and review text as 384-d embeddings	**pgvector**	Unstructured natural language that must be matched by meaning. Co-locating vectors with the catalog lets one query combine similarity with price/stock/rating filters and keeps vectors transactionally consistent with their rows. HNSW gives sub-10 ms search at 85k vectors (§9.3).
Product text as `tsvector`	**PostgreSQL full-text (GIN)**	Exact terms, brands and model numbers are better matched lexically; also the baseline the vector search is compared against.

A separate NoSQL store was deliberately **not** added: no part of the data is schema-less or graph-shaped enough to justify a second system to keep consistent. The one semi-structured field (`search_logs.filters` , `eval_runs.details`) uses PostgreSQL `jsonb` .

5. ER diagram and relational schema



Cardinality notes

Relationship	Cardinality	Implementation
User → Customer profile / Seller	1 : 0..1 each (role specialisation, disjoint)	`customer_profiles.user_id` PK+FK; `sellers (seller_id, role)` composite FK to `users (user_id, role)` with `CHECK (role = 'seller')`, so a seller row can only exist for a user whose role is `seller`
Category → Category	1 : N (self-referencing tree)	`categories.parent_id`; traversed with recursive CTEs
Seller → Product	1 : N	`products.seller_id` (ON DELETE RESTRICT)
Product → Embedding	1 : 0..1	separate table, so the model/version can change without touching the catalog
Cart ↔ Product	M : N	associative entity `cart_items` with composite PK
Order ↔ Product	M : N	associative entity `order_items` with composite PK (carries `quantity`, `unit_price`)
Order → Payment	1 : 1	`payments.order_id` UNIQUE

Relational schema (3NF)

Underlined = primary key, *italic* = foreign key.

```
users(user_id, email, password_hash, full_name, phone, role, is_active, created_at, last_login_at)
    UNIQUE(email), UNIQUE(user_id, role)

customer_profiles(user_id→users, date_of_birth, preferred_category_id→categories, taste_embedding,
taste_updated_at)

sellers(seller_id, role, store_name, gstin, support_email, is_verified, joined_at)
    FK (seller_id, role) → users(user_id, role),  CHECK (role = 'seller')

addresses(address_id, user_id→users, label, line1, line2, city, state, pincode, is_default)

categories(category_id, name, slug, parent_id→categories)

products(product_id, seller_id→sellers, category_id→categories, asin, title, brand, description,
features,
         price, mrp, stock_qty, reorder_threshold, image_url, avg_rating, review_count, is_active,
         created_at, updated_at, search_tsv)

product_embeddings(product_id→products, embedding, model, source_hash, embedded_at)

reviews(review_id, product_id→products, user_id→users, rating, title, body, verified_purchase,
        helpful_votes, created_at)
    UNIQUE(product_id, user_id)

review_embeddings(review_id→reviews, embedding, model, source_hash, embedded_at)

embedding_jobs(job_id, entity_type, entity_id, enqueued_at, started_at, processed_at, attempts,
last_error)

carts(cart_id, user_id→users)                        UNIQUE(user_id)
cart_items(cart_id→carts, product_id→products, quantity, added_at)           PK(cart_id, product_id)

orders(order_id, user_id→users, address_id→addresses, status, subtotal, shipping_fee, total_amount,
       placed_at, updated_at)
order_items(order_id→orders, product_id→products, quantity, unit_price, line_total)   PK(order_id,
product_id)
payments(payment_id, order_id→orders, method, amount, status, transaction_ref, created_at)
UNIQUE(order_id)
order_status_history(history_id, order_id→orders, old_status, new_status, changed_by→users,
changed_at)

stock_alerts(alert_id, product_id→products, stock_qty, threshold, created_at, resolved_at)
search_logs(log_id, user_id→users, query, mode, filters, result_count, latency_ms, created_at)
eval_runs(run_id, mode, model, k, query_count, precision_at_k, recall_at_k, mrr, ndcg_at_k,
          avg_latency_ms, details, run_at)
```

Normalisation argument

- **1NF**: every column is atomic. The only multi-valued attribute, product feature bullets, is stored as newline-separated text because it is only ever read and embedded as a unit, never queried per bullet.
- **2NF**: in the two tables with composite keys (`cart_items`, `order_items`), every non-key attribute depends on the whole key. `unit_price` depends on (order, product): it is a historical snapshot, deliberately *not* a copy of `products.price`.
- **3NF**: no transitive dependencies. Seller details live in `sellers`, category names in `categories`, delivery address in `addresses` (orders reference it, they do not copy it).
- **Deliberate, controlled denormalisation** (documented trade-offs, each kept consistent by the database itself):
  - `products.avg_rating`, `products.review_count`: recomputed by statement-level triggers with transition tables;
  - `order_items.line_total`, `products.search_tsv`: `GENERATED ALWAYS … STORED` columns;
  - `orders.subtotal / total_amount`: guarded by `CHECK (total_amount = subtotal + shipping_fee)`;
  - `customer_profiles.taste_embedding`: derived cache of purchase history, refreshed by `refresh_taste_profile()`.

6. Database implementation

Source: `db/01…07_*.sql`, applied in order by `npm run db:setup`.

6.1 DDL: tables, constraints, domains, indexes

- **19 tables**, 5 enumerated types, 5 validation **domains** (`email_address`, `money_inr`, `indian_pincode`, `indian_phone`, `gstin_code`) so the same rule applies everywhere a value is stored.
- **Keys:** surrogate identity primary keys, composite primary keys on the two associative tables (`cart_items`, `order_items`), `UNIQUE` on natural keys (email, store name, GSTIN, ASIN, one review per user per product).
- **Referential actions chosen deliberately:** `RESTRICT` where history must be preserved (orders → users, order items → products), `CASCADE` for owned data (cart items, embeddings), `SET NULL` where a record should survive its author (reviews).
- **Integrity beyond column checks:** `CHECK (mrp >= price)`, `CHECK (total_amount = subtotal + shipping_fee)`, a **composite foreign key** `sellers(seller_id, role) → users(user_id, role)` with `CHECK (role='seller')` so a seller row can only belong to a user whose role is seller, **partial unique indexes** (one default address per user, one open stock alert per product, one pending embedding job per entity), and generated columns (`line_total`, `search_tsv`).
- **Indexes:** B-tree (foreign keys, composite `(user_id, placed_at DESC)`), partial, **GIN** on the weighted `tsvector` and a trigram index, **HNSW** (`vector_cosine_ops`) on both embedding tables.
- **DDL evolution** (`ALTER`, `DROP`): `db/demo/ddl_demo.sql` walks through every DDL form on a scratch table; `npm run db:setup -- --reset` drops and rebuilds the schema.

6.2 Advanced features implemented

Concept	Where
Transactions & locking	`checkout()`: atomic, `SELECT … FOR UPDATE` in primary-key order (deadlock-free); `cancel_order()`
Stored functions / procedures	32 routines, e.g. `checkout`, `cancel_order`, `search_products`, `register_*`, `authenticate`, `recommend_for_me`, `restock_product` (PROCEDURE), `refresh_analytics` (PROCEDURE)
Triggers (15)	low-stock alerts (+ `NOTIFY`); embedding queue (de-duplicated); verified-purchase flag; rating aggregates (**statement-level with transition tables**); order **state-machine** guard; status audit trail; cart stock validation; email normalisation; `updated_at`
Views / materialized views	5 views (`v_product_cards`, `v_public_reviews`, `v_sales_lines`, `v_low_stock`, `v_category_tree`) and 2 materialized views refreshed `CONCURRENTLY` (`mv_daily_sales`, `mv_product_pairs`)
Indexing	B-tree, partial, GIN (FTS + trigram), HNSW
Full-text search	weighted `tsvector` (title A, brand B, features C, description D) + `ts_rank_cd`
Vector similarity search	cosine KNN, hybrid RRF, similar products, personalised recommendations

Concept	Where
Window functions & ROLLUP	`LAG`, moving averages, `RANK() OVER (PARTITION BY …)`, `GROUP BY ROLLUP`
Recursive CTE	category tree / breadcrumbs
Row-Level Security	13 tables, per-role policies
Security	`pgcrypto` bcrypt, `SECURITY DEFINER` with pinned `search_path`, column-level `GRANT`, least-privilege role

6.3 Triggers in detail

Trigger	Event	Behaviour
`trg_products_stock_alert`	AFTER INSERT/UPDATE of stock, threshold	opens/updates an alert when `stock_qty < reorder_threshold`, resolves it on restock; `pg_notify`
`trg_*_enqueue_embedding_*`	AFTER INSERT / UPDATE of embedded text	inserts one pending job (`ON CONFLICT DO NOTHING`), notifies worker; price/stock edits do **not** enqueue
`trg_reviews_verify`	BEFORE INSERT on reviews	sets `verified_purchase` from order history; ignores any client-supplied value
`trg_reviews_stats_*`	AFTER INSERT/UPDATE/DELETE, **FOR EACH STATEMENT** with transition tables	recomputes `avg_rating` / `review_count` once per affected product per statement
`trg_orders_status_guard`	BEFORE UPDATE of status	allows only pending→paid/cancelled, paid→shipped/cancelled, shipped→delivered; else `SS004`
`trg_orders_status_history`	AFTER INSERT/UPDATE of status	audit row with actor
`trg_cart_items_validate`	BEFORE INSERT/UPDATE	rejects quantity above stock or inactive products

Trigger functions that write to tables the application role cannot touch are `SECURITY DEFINER`. This was found and fixed through the test suite (§11): with invoker rights, restocking a product failed with *permission denied*.

6.4 Authentication and authorisation

- Registration and login are database functions (`register_customer`, `register_seller`, `authenticate`) that hash with bcrypt (`crypt(…, gen_salt('bf'))`). The application never sees a hash.
- Three roles: **customer, seller, admin**. Enforced at three layers: Next.js middleware (friendly redirect), route handlers (`requireRole`), and **Row-Level Security** (the authority).
- **RLS in practice:** one SQL statement `SELECT … FROM v_sales_lines` returns a seller's own lines for a seller, only a customer's own orders for a customer, everything for an admin, nothing for an anonymous visitor. Sellers see only *their* lines even inside a shared order. The seller dashboard queries contain no `WHERE seller_id = …`.

7. Vector database design

(Full design: `docs/vector-database-design.md`.)

Question (rubric §8)	Answer
What is converted to embeddings?	Product text (title, brand, category path, feature bullets, description) and review text (title + body)
Which model?	`all-MiniLM-L6-v2` (22M parameters), 384-d, mean-pooled and L2-normalised, run locally with transformers.js (ONNX, CPU); input capped at the model's 256-token training length
How are vectors stored?	`vector(384)` columns in `product_embeddings` and `review_embeddings` (separate from the catalog), plus `customer_profiles.taste_embedding` (mean vector of purchased products). Each row stores the model name and an md5 of the source text
Similarity metric?	Cosine distance (`<=>`), HNSW index `(m=16, ef_construction=64)`, `ef_search=200`, `iterative_scan` for filtered queries
How is relevance retrieved?	**Hybrid Reciprocal Rank Fusion** of three ranked lists (keyword, product vectors, best-matching review vector), filtered by relational predicates

7.1 Hybrid retrieval

```
score(p) = w_lex/(60 + rank_lexical) + 1/(60 + rank_semantic) + 0.5/(60 + rank_review)
w_lex    = 1.0 if some eligible product contains ALL query terms (rank only those)
         = 0.25 otherwise (OR-fallback; a partial keyword match is weak evidence)
```

7.2 Embedding consistency

Triggers enqueue a job whenever embedded text changes; a worker using `LISTEN/NOTIFY` and `FOR UPDATE SKIP LOCKED` embeds in batches and skips unchanged text. A re-embedding of 17,117 items took about 31 minutes on a laptop CPU; a single edit becomes searchable in seconds.

8. Important queries

See `docs/important-queries.md` for the annotated SQL. Highlights: hybrid RRF search (Q1); association rules with support, confidence and lift (Q2); month-over-month growth and moving averages (Q3, Q4); `ROLLUP` (Q5); rank within category (Q6, Q7); atomic checkout (Q8); low-stock trigger (Q9); vector recommendations with `LATERAL` (Q10, Q11); recursive CTE (Q12); trigram suggestions (Q13).

9. Results

9.1 Search quality

Method. Two labelled query sets: `eval/queries.json` (40 queries: 28 natural-language, 12 exact) and `eval/holdout.json` (18 queries, written *before* the fusion algorithm was revised and never used to tune it). A product is relevant if it lies in the query's category and its text matches a concept pattern; grade 2 if the concept is in the title, 1 if elsewhere. Metrics at k = 10; nDCG uses the graded labels. Labels are computed in SQL and are independent of the retrieval code. Natural-language queries deliberately use different words from the labels.

Result: main set (40 queries), final algorithm

Mode	nDCG@10	P@10	Recall@10	MRR	nDCG natural-language	nDCG exact
Keyword (full-text)	0.492	0.472	0.472	0.813	0.338	0.852
Semantic (vectors)	0.660	0.677	0.677	0.855	0.575	0.857
Hybrid (RRF)	**0.687**	**0.710**	**0.710**	**0.880**	**0.600**	**0.890**

Result: held-out set (18 queries), final algorithm

Mode	nDCG@10	P@10	MRR	nDCG natural-language	nDCG exact
Keyword	0.402	0.372	0.715	0.272	0.664
Semantic	0.733	0.733	0.889	0.714	0.771
Hybrid	**0.738**	**0.744**	0.861	**0.720**	0.773

What this shows, and what it does not.

- Vector search is the large win: nDCG 0.49 → 0.66 (main) and 0.40 → 0.73 (held-out) over keyword search, and it is what makes natural-language queries work (nDCG 0.34 → 0.58).
- **The first hybrid design was wrong, and we found out by measuring.** With equal-weight RRF, hybrid scored 0.649 (main) and 0.589 (held-out): *worse than semantic alone* (0.660 / 0.733). The per-query table showed why: for intent-style queries the OR-keyword list returned junk (matching words like "night" or "make"), and RRF gave it as much influence as the good semantic list. For *"make my backyard glow at night without wiring"*, hybrid scored 0.10 against 0.49 for semantic.
- **The fix** (conjunctive keyword match when one exists; otherwise OR at weight 0.25) lifted hybrid to 0.687 / 0.738. The weight 0.25 is a round constant chosen before looking at results; it was *not* fitted to the labels. Because it improved the **held-out** set too, the gain is not just overfitting to the 40 queries we had inspected.
- **Caveats.** Hybrid's lead over semantic is clear on the main set (+0.027 nDCG, +0.033 P@10) but within noise on the 18-query held-out set (+0.005), and semantic has a slightly higher MRR there. We therefore claim *hybrid is at least as good as semantic and far better than keyword*, with the clearest benefit on exact queries (0.890 vs 0.857), not that hybrid dominates. Labels are pattern-based

approximations of human judgement, and the catalog is 3,000 products, so absolute values are not comparable with other benchmarks.
- Before the revision the same runs are stored in `eval_runs` as variant `rrf-equal-weights`; the final ones as `rrf-adaptive`.

9.2 Application behaviour

Association rules: 3,840 product pairs; "frequently bought together" shown where lift > 1. Seed data contains planted co-purchase structure and seasonality (festive peaks), so trends and rules are meaningful but **synthetic**.

9.3 Index performance (`npm run bench`, `EXPLAIN ANALYZE`)

Run: 2026-10-27T14:56:28.043Z · PostgreSQL 17.11 (Debian 17.11-1.pgdg12+2) · pgvector 0.8.7 Medians over 20 queries; every figure is the Execution Time reported by EXPLAIN (ANALYZE).

1. Vector similarity (top-10 by cosine distance)

Table	Vectors	Exact scan	HNSW (ef_search 40)	HNSW (ef_search 200)
product_embeddings (real)	3,001	3.05 ms	1.05 ms (recall 1.000)	3.52 ms (recall 1.000)
review_embeddings (real)	14,116	16.00 ms	17.11 ms (recall 0.995)	3.91 ms (recall 1.000)
scaled-up (real + noisy copies)	85,585	88.54 ms	6.16 ms (recall 0.950)	9.95 ms (recall 1.000)

HNSW build on the scaled-up set: 120.5 s · index 167 MB vs heap 134 MB.

2. Full-text search (GIN on weighted tsvector)

Query term(s)	Matches	Seq scan	GIN index	Speed-up
headphones	28	17.35 ms	0.17 ms	105.2×
dog toy	29	16.57 ms	0.20 ms	82.0×
stainless steel	113	17.81 ms	0.32 ms	56.2×
waterproof	199	18.10 ms	0.49 ms	36.9×
yoga	67	19.07 ms	0.23 ms	82.9×

3. B-tree composite index `orders (user_id, placed_at DESC)`

Query	Seq scan + sort	B-tree index	Speed-up
latest 20 orders of one customer (15,417 orders)	2.23 ms	0.13 ms	17.0×

4. Materialized view vs live aggregation

Monthly revenue report	Live join + aggregate	Materialized view	Speed-up
platform-wide	63.10 ms	19.75 ms	3.2×

Reading the vector table honestly. At 3k–14k vectors an exact scan already takes only 3–18 ms, so HNSW is not meaningfully faster there, and differences of a few milliseconds are within run-to-run noise (random probes, small medians). The index pays off at scale: with 85,585 vectors an exact scan costs about 80–90 ms against 4–10 ms for HNSW at recall 0.95–1.00. The GIN index gives 30–110× over a sequential scan and the composite B-tree 24×. The materialized view is 3.7× faster than the live aggregate, at the price of staleness until refreshed.

9.4 Concurrency lab

Two buyers race for the last unit (T2 starts 300 ms after T1). **Naive read-then-write** (no lock): both read stock = 1, both sell, stock ends at 0 with **2 units sold**: a lost-update anomaly. `checkout()` (row lock): T2 blocks until T1 commits, then reads stock = 0 and fails with `SS001`: **1 unit sold**. Reproduce with `npm run race` or at `/admin/lab`. The automated test `concurrency: two buyers racing for the last unit` asserts the outcome.

10. Application screenshots

ShopSense | Try "gift for a dad who loves gardening" or "quiet keyboard for night typing" | Hybrid | Search | Sign in | Create account

Describe what you need. We'll find it.

Hybrid search fuses PostgreSQL full-text ranking with vector similarity over product descriptions and real customer reviews, so plain-language searches work.

Search

gift for a dad who loves gardening | something to keep my dog's teeth clean | comfortable shoes for standing all day | keep coffee hot for hours | toy that teaches kids to code | Nintendo Switch controller

Shop by category

- Beauty and Personal Care — 375 products
- Clothing Shoes and Jewelry — 375 products
- Electronics — 376 products
- Home and Kitchen — 375 products
- Patio Lawn and Garden — 375 products
- Pet Supplies — 375 products
- Sports and Outdoors — 375 products
- Toys and Games — 375 products

Trending this month

Most units sold in the last 30 days



STEM TOYS
Playz Edible Poop! Food Science STEM Chemistry Kit - DIY Make...
★★★★★ (6)
₹2,899 ~~₹4,179~~ 31% off
12 sold recently



PRESCHOOL
MorisMos Giant Teddy Bear Stuffed Animals, Big Teddy Bea...
★★★★☆ (4)
₹5,729 ~~₹7,369~~ 22% off
11 sold recently



COMPUTERS & ACCESSORIES
Amazon Fire HD 10 Tablet Cover (Only compatible with 11th...
★★★☆☆ (6)
₹3,319 ~~₹4,969~~ 33% off
10 sold recently



DOGS
Furhaven Large Pillow Dog Bed Comfy Couch Plush & Decor...
★★★★☆ (6)
₹3,319 ~~₹4,619~~ 28% off
10 sold recently



GENERAL
Fire HD 8 Tablet (8" HD Display, 16 GB) - Black (Previous...
★★★★★ (6)
₹1,589 ~~₹2,319~~ 31% off
9 sold recently



HEADPHONES, EARBUDS & ACCESSORIES
AirPods Case, Full Protective Silicone AirPods Accessories...
★★★★★ (6)
₹909 ~~₹1,049~~ 13% off
9 sold recently



MEN
Crocs Unisex-Adult Classic Clog
★★★★★ (6)
₹1,829 ~~₹2,539~~ 28% off
9 sold recently



SKIN CARE
Hurraw! Raspberry Tinted Lip Balm: (Sheer Berry Tint) Organi...
★★★☆☆ (6)
₹739 ~~₹1,049~~ 30% off
9 sold recently



TOOLS & ACCESSORIES
KISS Looks So Natural Trio Lash Medium Combo Pack, False...
★★★★☆ (6)
₹609 ~~₹759~~ 20% off
9 sold recently



PUZZLES
GEMEM 155 Pcs Wooden Pattern Blocks Set Geometric Shape...
★★★★★ (4)
₹1,079 ~~₹1,309~~ 18% off
9 sold recently

Top-rated deals

Rated 4★ and above, sorted by discount



MEN
X XHtang Men's Ratchet Belt with Genuine Leather, Slide Bel...
★★★★★ (6)
₹1,579 ~~₹2,529~~ 38% off



TELEVISION & VIDEO
UGREEN HDMI Cable Right Angle 4K 90 Degree HDMI Cor...
★★★★★ (4)
₹749 ~~₹1,199~~ 38% off



WOMEN
Hey Dude Women's Wendy Scarlett Leopard Size 10 |...
★★★★☆ (6)
₹829 ~~₹1,309~~ 37% off



MEN'S GROOMING
Old Spice Lemon Lime, 3.0 oz
★★★★☆ (6)
₹259 ~~₹409~~ 37% off



WOMEN
Mishansha Women's Running Walking Shoes
★★★★☆ (6)
₹2,489 ~~₹3,969~~ 37% off



WOMEN



GARDENING & LAWN CARE



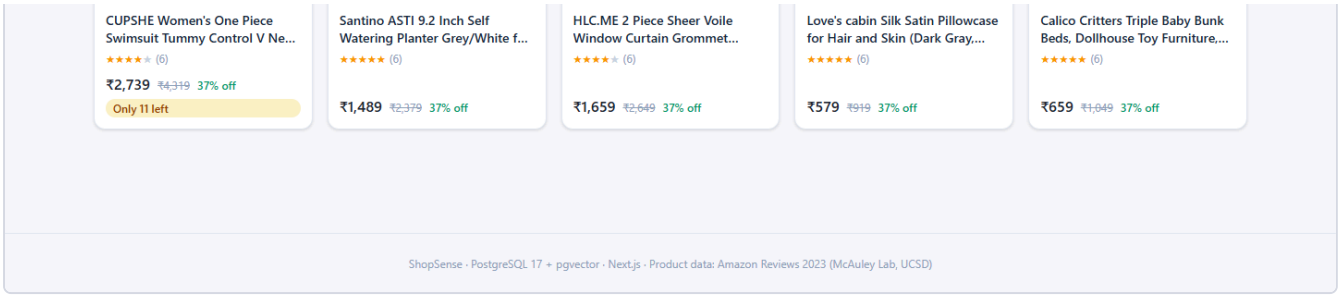
HOME DÉCOR PRODUCTS



BEDDING



DOLLS & ACCESSORIES



Home: natural-language search box, example queries, categories, trending products.

ShopSense | gift for a dad who loves gardening | Hybrid | Search | Sign in | Create account

CATEGORY: All categories

MIN ₹ | MAX ₹

MIN RATING: Any

☐ In stock only

Apply filters

Results for “gift for a dad who loves gardening”

Keyword | Semantic | Hybrid | Compare all 3

24 results · query embedding 1028 ms · SQL 1857 ms · Hybrid (RRF)



OUTDOOR DÉCOR

H&GT Garden Kneeler and Seat, Foldable Garden Stool Heavy...

★★★★★ (3)

₹4,399 ~~₹5,849~~ 25% off

keyword #1 | meaning #1



PRESCHOOL

Liberty Imports Pull Along Garden Wagon and Gardening...

★★★★☆ (4)

₹879 ~~₹1,089~~ 19% off

meaning #2 | review #5

"Bought this for my grand daughter, she is 4 and loves it ! She loves to help me in the garden, really fun for her."



OUTDOOR DÉCOR

Garden Gnome Statue, Resin Gnome Figurine Holding Flowe...

★★★★★ (3)

₹1,229 ~~₹1,699~~ 28% off

meaning #7 | review #12

"How cute is he? And he is so helpful, holding a leaf that you can fill with bird seeds. I love the placement of the lights...



GARDENING & LAWN CARE

TOREVSIOR 6Ft Vertical Garden Planter, Vertical Elevated Plant...

★★★★★ (4)

₹6,629 ~~₹7,009~~ 5% off

meaning #10 | review #33

"Vertical Garden Bed, 6Ft Vertical Elevated Planter for Plants, Patio Balcony Indoor and Outdoor (Metal)...



GARDENING & LAWN CARE

HOMY Garden Tool Set, Stainless Steel Heavy Duty Gardening Ki...

★★★★☆ (5)

₹609 ~~₹889~~ 31% off

meaning #15 | review #44

"HOMY Garden Tool Set, Stainless Steel Heavy Duty Gardening Kit with Soft Rubberized Non-Slip Ergonomic Handl...



GRILLS & OUTDOOR COOKING

N NOBLE FAMILY 21PCS Professional Stainless Steel Gril...

★★★★★ (3)

₹2,489 ~~₹2,899~~ 14% off

meaning #42 | review #1

"We love to camp so I got this for my daughters dad for Father's Day - gave it to him early & he loves it"



GARDENING & LAWN CARE

Outland Living 6-Ft Raised Garden Bed - Vertical Garden...

★★★★★ (4)

₹11,539 ~~₹13,199~~ 13% off

meaning #11 | review #90

"WE LOVED OUR VERTICAL GARDENS (2) SO MUCH, WE BOUGHT ONE FOR OUR DAUGHTER AND HER HUSBAND...



PRESCHOOL

Sago Brothers Air Dry Clay 24 Colors, Kids Toys Modeling Cla...

★★★★★ (6)

₹1,409 ~~₹2,179~~ 35% off

meaning #38 | review #17

"My daughter absolutely love this clay. She made so many creative projects from it and let them dry. She made...



GARDENING & LAWN CARE

Amazon Exclusive Star Wars The Child Chia Pet Floating Edition...

★★★★★ (5)

₹2,069 ~~₹3,059~~ 32% off

Only 5 left

meaning #17 | review #94

"Secret Santa gift. Packaged well, price under $20."



GARDENING & LAWN CARE

Classic Home and Garden 260014P-433 Piglet Planter,...

★★★★★ (3)

₹1,099 ~~₹1,599~~ 27% off

meaning #3



GARDENING & LAWN CARE

Pure Garden 50-LG1150 Stands – Set of 2 Indoor or Outdoor...

★★★★★ (5)

₹3,619 ~~₹4,049~~ 11% off

meaning #4



PRESCHOOL

TOMY John Deere Sit-N-Scoot Tractor Toy, Green, One Size

★★★★☆ (5)

₹919 ~~₹979~~ 6% off

meaning #33 | review #43

"Our grandson loved it and it makes a good photo prop for the studio. Nice for any child that enjoys farms."



OUTDOOR DÉCOR

Mosteck Anodized Aluminium Wind Chimes 26" Inches, Gift f...

★★★★☆ (6)

₹989 ~~₹1,479~~ 33% off

meaning #5



PUZZLES

Baby Personalized Wood Name Puzzle With Custom Design -...

★★★★★ (3)

₹1,409 ~~₹1,589~~ 11% off

meaning #6



OUTDOOR DÉCOR

HTOOR Old Man Tree Hugger, Garden Yard Art Decoration an...

★★★☆☆ (3)

₹2,539 ~~₹3,109~~ 18% off

meaning #8



GARDENING & LAWN CARE

EZIOSS Raised Garden Bed - Galvanized Planter Box for...

★★★★★ (3)

₹4,899 ~~₹7,349~~ 33% off

meaning #9



OUTDOOR DÉCOR

ESTTOP 2FT 24LT Led Lighted Birch Tree, 24" White Money...

★★★★★ (4)



PRESCHOOL

MEGA BLOKS Fisher-Price Toddler Building Blocks, Green...

★★★★★ (3)

₹699 ~~₹759~~ 8% off



GARDENING & LAWN CARE

Genuine Fred Fancy Ceramic Mini Air Plant/Succulent Holde...

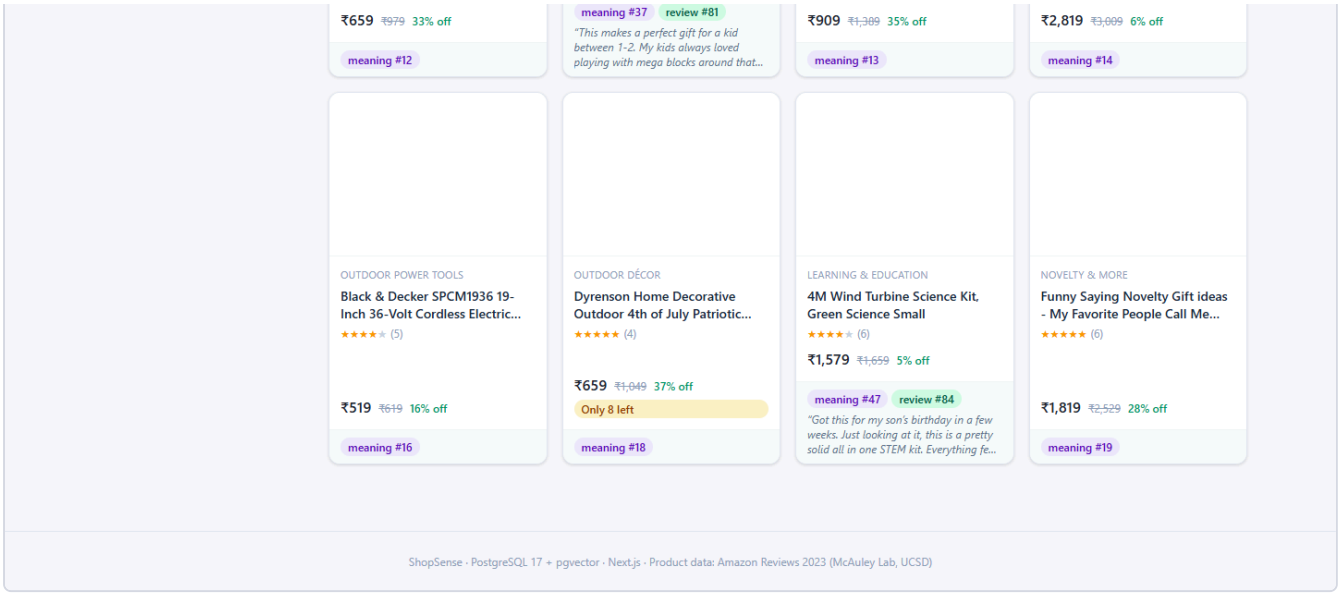
★★★★☆ (3)



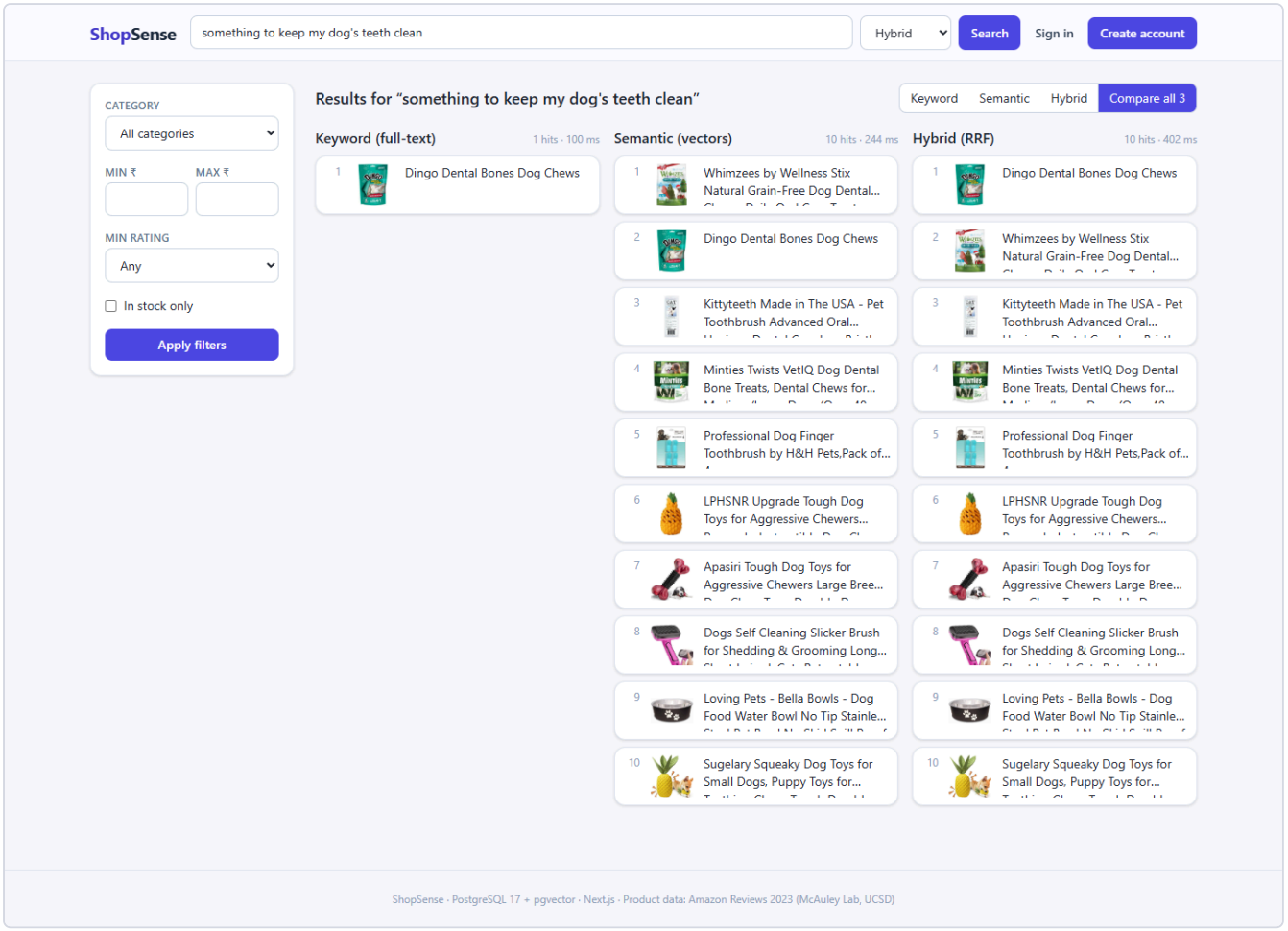
NOVELTY & MORE

Grumpy Old Man Killing It Funny Grandpa Sweatshirt

★★★★★ (6)



Hybrid search for "gift for a dad who loves gardening", with per-list ranks and the matching customer-review sentence.



"Compare all 3": keyword, semantic and hybrid results side by side.

ShopSense | comfortable shoes for standing all day | Hybrid | Search | Sign in | Create account

CATEGORY: All categories

MIN ₹ | MAX ₹: 3000

MIN RATING: 4★ & up

In stock only

Apply filters

Results for “comfortable shoes for standing all day”

Keyword | Semantic | Hybrid | Compare all 3

24 results · query embedding 12 ms · SQL 189 ms · Hybrid (RRF)



WOMEN

LongBay Women's Chenille Knit Bootie Slippers Cute Plush Flee...

★★★★★ (6)

₹2,319 ~~₹3,089~~ 25% off

keyword #1 | meaning #7 | review #22

“This is my second pair of these slippers. I am retired and home each day. I do not wear outside shoes in my home he...



SPORTS

BPS New Zealand 'Second Skin' Soft and Ultra Stretch Water...

★★★★☆ (4)

₹1,239 ~~₹1,959~~ 37% off

keyword #3 | meaning #24 | review #17

“Just what I need, and fit my barefoot fins.”



MEN

Teva Men's Mush II Flip-Flop

★★★★★ (6)

₹1,959 ~~₹2,109~~ 7% off

keyword #2 | meaning #38 | review #48

“Its not called the “mush” for nothing. Its does have a nice mushy feel to it when you walk, so its pretty comfortable on...



WOMEN

Women's Walking Shoes Sock Sneakers Mesh Slip On Air...

★★★★☆ (3)

₹2,989 ~~₹3,369~~ 11% off

meaning #2 | review #8

“Wonderful comfortable bought 2 more pairs! Great for wide feet!”



WOMEN

Women's Walking Shoes Sock Sneakers - Mesh Slip On Air...

★★★★☆ (6)

₹2,899 ~~₹4,029~~ 28% off

meaning #1 | review #13

“I really like these sneakers. They are comfortable and look good but they are definitely not for walking. I wore them...



MEN

KUBUA Slippers for Men and Womens Indoor House Shoes...

★★★★☆ (5)

₹2,989 ~~₹3,619~~ 17% off

meaning #6 | review #9

“I have tried many options for house shoes, I finally have found comfort, price. I can't wait to order another pair....



MEN

ASICS Men's JOLT 2 Shoes

★★★★★ (3)

₹669 ~~₹789~~ 15% off

meaning #3 | review #21

“They fit well, are comfortable and the price was quite good. I am happy.”



MEN

Crocs Unisex-Adult Classic Clog

★★★★★ (6)

₹1,829 ~~₹2,539~~ 28% off

meaning #15 | review #2

“I have 8 pair now so I must I really like them”



WOMEN

HEAWISH Women's Floral Ballet Flats for Women Black Beige Sl...

★★★★☆ (3)

₹2,239 ~~₹3,159~~ 29% off

meaning #8 | review #16

“These are comfortable shoes – unless you have to wear them for hours on a hard floor, then they might not be quit...



WOMEN

Hey Dude Women's Wendy Scarlett Leopard Size 10 |...

★★★★☆ (6)

₹829 ~~₹1,309~~ 37% off

meaning #11 | review #10

“One of the most comfortable pairs of shoes I own. I would size down as they run bigger but easy to care for and I...



WOMEN

Skechers Performance Women's Go Walk 4 Propel Walking Shoe

★★★★☆ (5)

₹1,589 ~~₹1,819~~ 13% off

meaning #14 | review #5

“These shoes are so comfortable. They look great as well.”



WOMEN

DADAWEN Women's Classic T-Strap Platform Mid-Heel Squar...

★★★★☆ (6)

₹2,839 ~~₹3,289~~ 14% off

meaning #9 | review #18

“These are super comfy, and easy to walk in for a heel. The insoles are squishy but a little thin, I can feel...



MEN

Under Armour Men's Charged Assert 9 Running Shoe

★★★★★ (6)

₹1,669 ~~₹2,379~~ 30% off

meaning #17 | review #4

“5 stars I gave it 4 stars under workout because, I use them as work shoes so I don't know. They are comfortable &...



WOMEN

DOUSSPRT Women's Water Shoes Quick Drying Sports Aq...

★★★★☆ (6)

₹2,239 ~~₹2,789~~ 20% off

meaning #10 | review #20

“I bought these shoes for a trip to Turkey along the sea of turquoise. Didn't really know how much I would actually wear...



WOMEN

Women's Canvas Shoes Fashion Sneakers Low Top Tennis Shoes...

★★★★☆ (5)

₹1,659 ~~₹2,219~~ 25% off

meaning #5 | review #38

“As of this review I don't know how much these sneakers cost but hopefully not more than $30.00. I received a size 11...



WOMEN

Skechers Performance Women's Go Walk Move Solstice Flip Flop

★★★★★ (4)

₹2,899 ~~₹3,979~~ 27% off

Only 7 left

meaning #19 | review #6

“I LOVE THESE SHOES!!! I have had several pairs. Sooo comfortable!! Great for summer”



WOMEN

FUNKYMONKEY Women's



BOOT SHOP

DailyShoes Women's Military



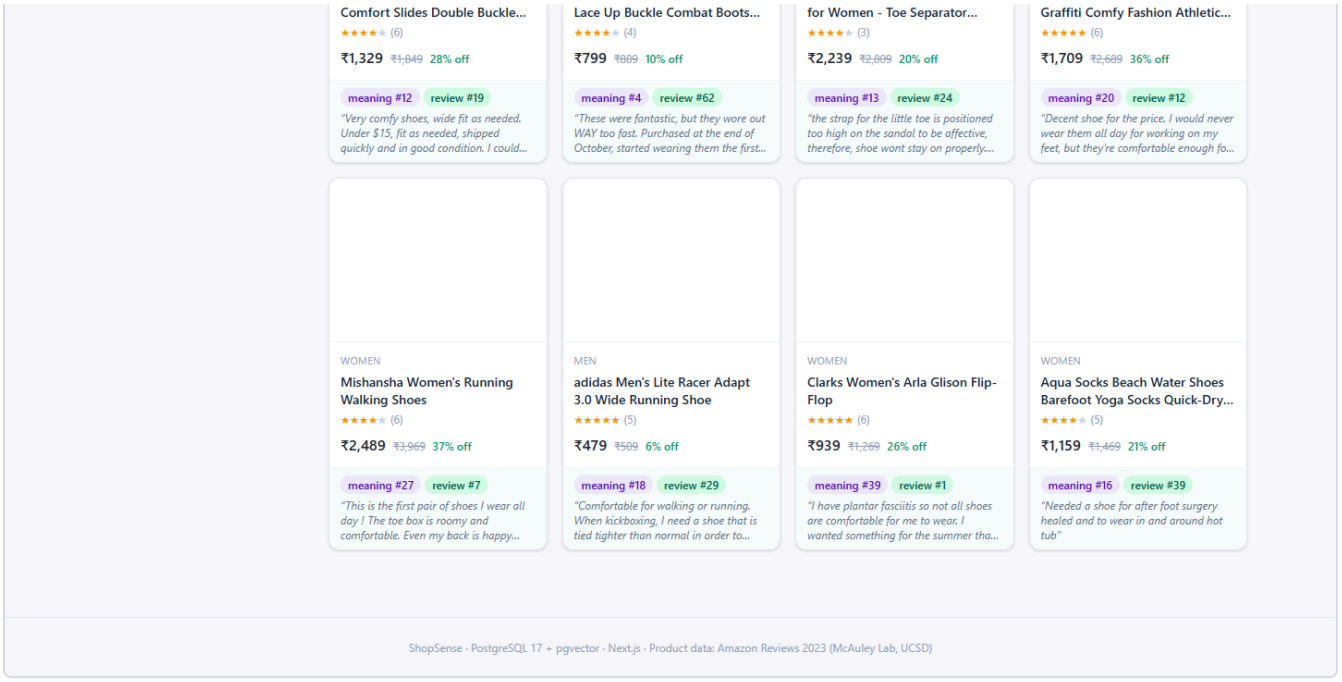
FOOT, HAND & NAIL CARE

Pedi Couture Pedicure Sandals



TENNIS & RACKET

BRONAX Men's Tennis Shoes



Relational filters (price, rating, in stock) applied inside the same SQL function.

ShopSense | Try "gift for a dad who loves gardening" or "quiet keyboard for night typing" | Hybrid | Search | Sign in | Create account

Electronics > Computers & Accessories



Wireless Keyboard and Mouse Combo, EDJO Full-Sized Ergonomic Computer Keyboard with Wrist Rest and 3 Level DPI Adjustable Wireless Mouse for Windows, Mac OS Desktop/Laptop/PC (White)

by **EDJO** · sold by **VoltKart Electronics**

★★★★★ 4.5 · 6 reviews

₹2,899 ~~₹3,779~~ 23% off

Sign in to buy

- 【ERGONOMIC WIRELESS KEYBOARD AND MOUSE COMBO】EDJO Full-sized wireless keyboard is ergonomically designed with Palm Rest and folding holder that can keep it at an optimum slope， prevent your wrists from hurting while long sessions of typing. Keyboard is also designed with anti-slide pads so it will stay in place when you're typing quickly. NOTE: THE USB RECEIVER IS INSERTED IN THE BATTERY COMPARTMENT OF THE MOUSE, YOU CAN FIND IT WHEN OPEN THE MOUSE BATTERY COVER.
- 【PLUG & PLAY 2.4G WIRELESS CONNECTION】One 2.4 GHz USB receiver can connect both the keyboard and mouse, can also be used separately, plug & play, no need to download any software. 2.4 GHz wireless provides a powerful and reliable connection up to 33 feet(10m) without any delays.
- 【AUTOMATIC POWER SAVING FUNCTION】The EDJO wireless keyboard and mouse combo has the function of automatically entering the power-saving state. When you stop using the keyboard more than 30 minutes, stop using the mouse more than 25 seconds, they will enter sleep mode respectively to save power. This feature greatly extends the battery life, you can click any button to activity the device. The keyboard need 1 x AA battery, the mouse need 1 x AA battery. (Battery Not Included)
- 【WIRELESS OPTICAL MOUSE】The mouse was optical design, even on some smooth surfaces, precise control can be obtained. 3-level adjustable DPI (800/1200/1600) allows you to choose your favorite moving speed of cursor. In addition, this wireless mouse is symmetry design, no matter you are Right handed Left handed, both hands are available, suitable for all people.
- 【UNIVERSAL COMPATIBILITY & AFTER-SALES SERVICE】This wireless keyboard and mice combo compatible with windows XP/Vista/7/8/10/X, Mac and other operating system. Works well with desktops, Chrome-book, PC, Laptop, Computer and more. If you encounter any problems during use, please contact us via Amazon email, we will provide a satisfactory solution.

Frequently bought together

Association rules mined from order history (materialized view `mv_product_pairs`). Confidence = P(this item | viewed item); lift > 1 means bought together more often than chance.



COMPUTERS & ACCESSORIES

HOTWEEMS Wireless D-09 Computer Mouse USB Cordless Mice for Laptop, Er...

★★★★☆ (6)

₹909 ~~₹1,269~~ 28% off

conf 33% · lift 492.7× · 5 orders



COMPUTERS & ACCESSORIES

Logitech M585 Multi-Device Wireless Mouse – Control and Move...

★★★★★ (4)

₹2,389 ~~₹3,049~~ 22% off

conf 20% · lift 128.5× · 3 orders

Similar products

Nearest neighbours in embedding space (cosine similarity, HNSW index)



COMPUTERS & ACCESSORIES

HOTWEEMS Wireless D-09 Computer Mouse USB Cordless Mice for Laptop, Er...

★★★★☆ (6)

₹909 ~~₹1,269~~ 28% off

similarity 0.68



COMPUTERS & ACCESSORIES

Microsoft Arc Wireless Keyboard for PC and Xbox 360 - Black

★★★★★ (3)

₹1,169 ~~₹1,629~~ 28% off

Only 2 left

similarity 0.62



COMPUTERS & ACCESSORIES

Logitech M585 Multi-Device Wireless Mouse – Control and Move...

★★★★★ (4)

₹2,389 ~~₹3,049~~ 22% off

similarity 0.57



COMPUTERS & ACCESSORIES

Verbatim Slimline Wired Keyboard USB Plug-and-Play Numeric Keypad Adjustab...

★★★★☆ (6)

₹829 ~~₹1,019~~ 19% off

similarity 0.55



COMPUTERS & ACCESSORIES

ELECOM Relacon Handheld Trackball Mouse, Thumb Control, 2.4GHz Wireless,...

★★☆☆☆ (3)

₹4,559 ~~₹4,829~~ 6% off



COMPUTERS & ACCESSORIES

mafiti RK101 Computer Keyboard Mouse Combo Wired, RGB Backlit USB Keyboard...

★★★☆☆ (6)

₹1,659 ~~₹1,859~~ 11% off



GENERAL

Apple Magic Mouse: Wireless, Bluetooth, Rechargeable. Works with Mac or iPad;...

★★★★☆ (6)

₹6,559 ~~₹8,539~~ 23% off

Only 6 left

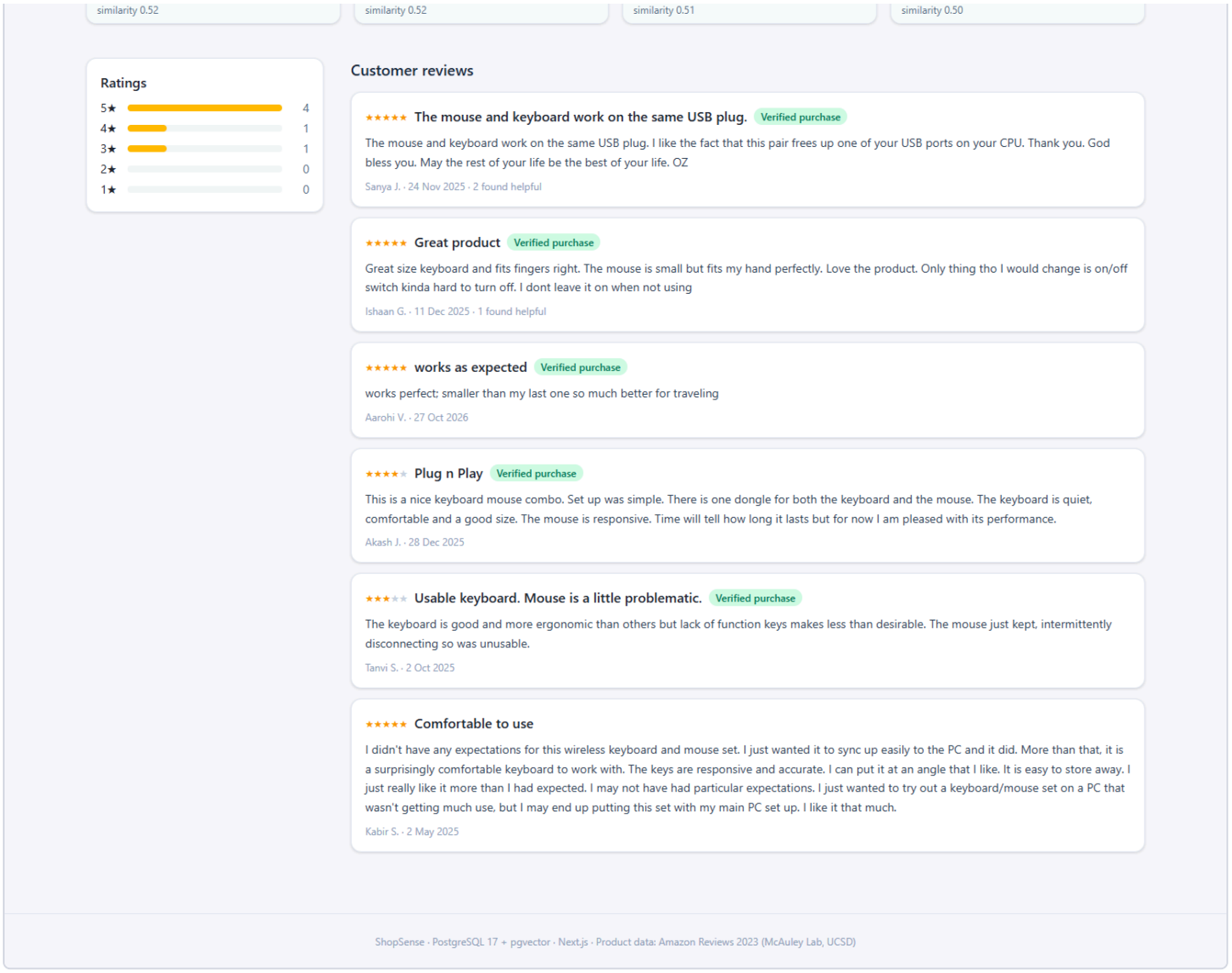


COMPUTERS & ACCESSORIES

BFRIENDit Wired USB Keyboard with Cover, Comfortable Quiet Chocolate Keys...

★★★★★ (4)

₹2,399 ~~₹3,719~~ 35% off



Product page: ratings, reviews, similar products (vector KNN), frequently bought together (association rules).

ShopSense | Try "gift for a dad who loves gardening" or "quiet keyboard for night typing" | Hybrid | Search | Orders | Cart | Aarav Sharma · Customer | Sign out

Describe what you need. We'll find it.

Hybrid search fuses PostgreSQL full-text ranking with vector similarity over product descriptions and real customer reviews, so plain-language searches work.

Search

gift for a dad who loves gardening | something to keep my dog's teeth clean | comfortable shoes for standing all day | keep coffee hot for hours | toy that teaches kids to code | Nintendo Switch controller

Recommended for you

Nearest neighbours of your taste vector: the average embedding of everything you have bought



COMPUTERS & ACCESSORIES
Amazon Basics Double Nylon Braided USB A Cable with...
★★★★★ (4)
₹2,289 ~~₹2,639~~ 13% off
match 68%



WEARABLE TECHNOLOGY
Tobfit 4 Pack Bands Compatible for Fitbit Charge 4 / Fitbit...
★★★★★ (3)
₹749 ~~₹969~~ 23% off
match 64%



SPORTS
Syncwire Waterproof Pouch Bag with Adjustable Waist Strap (2...
★★★★☆ (5)
₹1,489 ~~₹1,579~~ 6% off
match 63%



OUTDOOR RECREATION
Frelaxy Hi-Visibility Backpack Rain Cover with Reflective Strip...
★★★★★ (3)
₹1,079 ~~₹1,309~~ 18% off
match 63%



COMPUTERS & ACCESSORIES
LOVEVOOK Geometric Luminous Purses and Handbags for...
★★★★☆ (6)
₹3,149 ~~₹4,349~~ 28% off
Only 1 left
match 62%



SPORTS
HUGE SPORTS Men's Splice UV Sun Protection UPF 50+ Skins...
★★★★★ (4)
₹2,159 ~~₹2,329~~ 7% off
match 61%



WEARABLE TECHNOLOGY
Maledan Bands Compatible with Fitbit Charge 3 and Charge 4,...
★★★★★ (5)
₹749 ~~₹1,029~~ 27% off
match 61%



WEARABLE TECHNOLOGY
Tobfit Bands Compatible with Fitbit Versa 2 and Fitbit...
★★★★★ (3)
₹499 ~~₹779~~ 36% off
match 60%



MEN
Amazon Essentials Men's Lightweight Water-Resistant...
★★★★☆ (3)
₹1,879 ~~₹2,559~~ 27% off
match 59%



WEARABLE TECHNOLOGY
Tobfit 4 Pack Bands Compatible with Fitbit Alta/Alta HR Bands,...
★★★★★ (6)
₹829 ~~₹889~~ 7% off
match 59%

Shop by category

- Beauty and Personal Care — 375 products
- Clothing Shoes and Jewelry — 375 products
- Electronics — 376 products
- Home and Kitchen — 375 products
- Patio Lawn and Garden — 375 products
- Pet Supplies — 375 products
- Sports and Outdoors — 375 products
- Toys and Games — 375 products

Trending this month

Most units sold in the last 30 days



STEM TOYS
Playz Edible Poop! Food Science STEM Chemistry Kit - DIY Make...
★★★★★ (6)
₹2,899 ~~₹4,179~~ 31% off
12 sold recently



PRESCHOOL
MorisMos Giant Teddy Bear Stuffed Animals, Big Teddy Bea...
★★★★☆ (4)
₹5,729 ~~₹7,369~~ 22% off
11 sold recently



COMPUTERS & ACCESSORIES
Amazon Fire HD 10 Tablet Cover (Only compatible with 11th...
★★★☆☆ (6)
₹3,319 ~~₹4,969~~ 33% off
10 sold recently



DOGS
Furhaven Large Pillow Dog Bed Comfy Couch Plush & Decor...
★★★★☆ (6)
₹3,319 ~~₹4,619~~ 28% off
10 sold recently



GENERAL
Fire HD 8 Tablet (8" HD Display, 16 GB) - Black (Previous...
★★★★★ (6)
₹1,589 ~~₹2,319~~ 31% off
9 sold recently

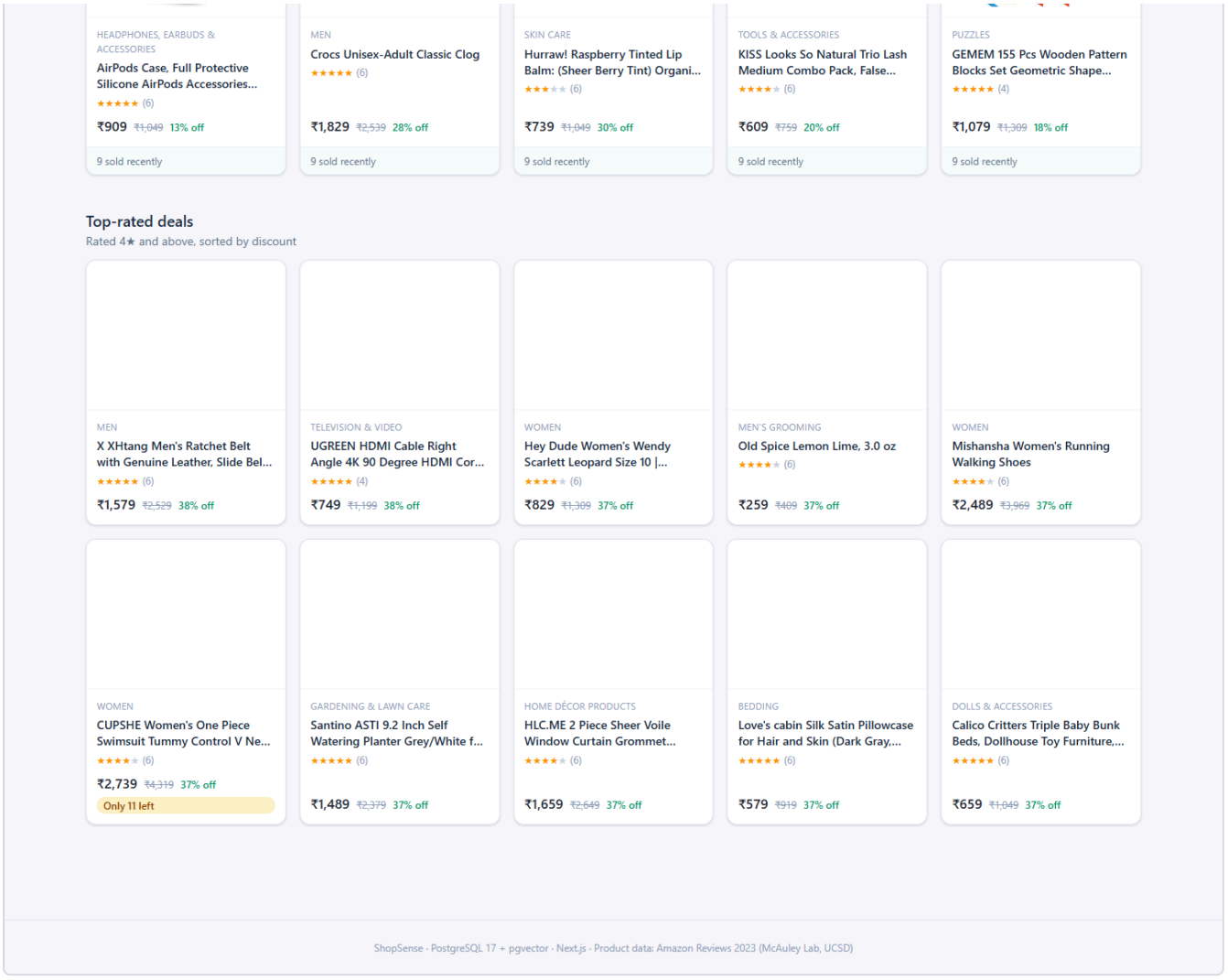




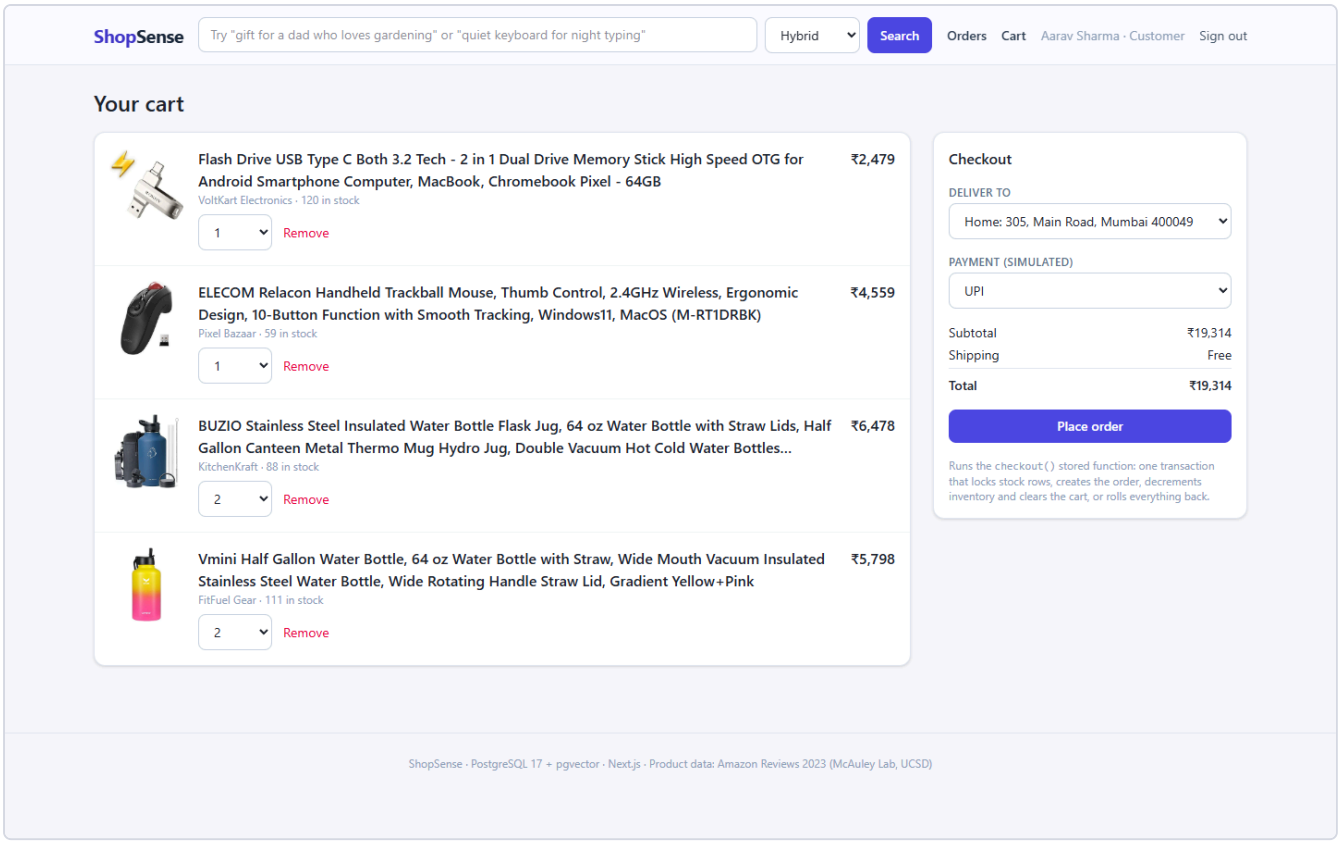








Signed-in customer: recommendations from the taste vector.



Cart and checkout (one atomic stored function).

ShopSense | Try "gift for a dad who loves gardening" or "quiet keyboard for night typing" | Hybrid | Search | Orders | Cart | Aarav Sharma · Customer | Sign out

Your orders

Order #15336 Shipped 25 Oct, 02:37 pm · UPI (success) · to Mumbai 400049 — **₹10,167**


Ultra Game NFL Mens Raglan Baseball 3/4 Long Sleeve T-Shirt × 1 ₹1,659


Franklin Sports Badminton Set - Portable Badminton Set - Adult and Kids Badminton Net - Perfect Backyard/Lawn Game - Includes 4 Badminton Racquets - Intermedia... × 1 ₹4,719


Franklin Sports NCAA Kids Football Uniform Set - NFL Youth Football Costume for Boys & Girls - Set Includes Helmet, Jersey & Pants × 1 ₹3,789

Shipped 25 Oct, 02:37 pm

Order #15041 Shipped 20 Oct, 09:22 am · COD (pending) · to Mumbai 400049 — **₹4,767**


60ft (18.3M) High Speed HDMI Cable Male to Male with Ethernet Black (60 Feet/18.3 Meters) Supports 4K 30Hz, 3D, 1080p and Audio Return CNE620107 × 1 ₹3,149


Cablecreation RCA Audio Cable 3.5mm Male to 2-Male RCA Cable, Y Splitter Design Stereo Audio RCA Male Cable, 3 Meters, Black × 2 ₹1,618

Shipped 20 Oct, 09:22 am

Order #14412 Delivered 9 Oct, 12:39 pm · UPI (success) · to Mumbai 400049 — **₹4,897**


Ultra Clarity Cables Coaxial Cable 20ft (2 Pack) - Triple Shielded RG6 Coax TV Cable Cord in-Wall Rated Gold Plated Connectors Digital Audio Video with Male F Conne... × 1 ₹1,659


FosPower (3 Feet 24K Gold Plated Toslink Digital Optical Audio Cable (S/PDIF) - [Zero RFI & EMI Interference] Metal Connectors & Ultra Durable Nylon Braided Jacket × 1 ₹749


Cmple - Optical Audio Cable 100ft - Slim Fiber Optic Cable Ultra-Thin Gold Plated Toslink Cable Digital S/PDIF Audio for Soundbar Av Receivers, TV Sound Bar, Game ... × 1 ₹2,489

Delivered 9 Oct, 12:39 pm

Order #13733 Cancelled 26 Sept, 11:55 pm · CARD (refunded) · to Mumbai 400049 — **₹4,888**


DAVID ARCHY 3 Pack Men's Ultra Soft Mesh Quick Dry Sports Underwear Breathable Boxer Briefs × 1 ₹2,819


5Mayi Men's Underwear Boxer Briefs Cotton Regular Long Mens Boxer Briefs Underwear Men Pack S M L XL XXL × 1 ₹2,069

Cancelled 26 Sept, 11:55 pm

Order #13719 Delivered 26 Sept, 11:37 pm · UPI (success) · to Mumbai 400049 — **₹2,158**


BASSDASH UPF 50+ UV Sun Protection Neck Gaiter Fishing Mask Hunting Kayaking Hiking Cycling Ski Sports × 1 ₹829


KALILY 12pcs/9pcs/6pcs Headband Bandana - Face Mask Headwear Neck Gaiter Shield Scarf × 1 ₹1,329

Delivered 26 Sept, 11:37 pm

Order #13618 Delivered 23 Sept, 09:29 am · CARD (success) · to Mumbai 400049 — **₹1,557**


Shintop Cord Organizer, 6 Pack Black Charger Holder Cable Clips Wire Clips for Organizing Desk Nightstand Car Phone Charger × 1 ₹499


SOULWIT Cable Holder Clips, 3-Pack Cable Management Cord Organizer Clips Silicone Self Adhesive for Desktop USB Charging Cable Nightstand Power Cord Mouse ... × 2 ₹1,058

Delivered 23 Sept, 09:29 am

Order #13520 Delivered 21 Sept, 11:05 am · UPI (success) · to Mumbai 400049 — **₹999**


Cat6 Ethernet Cable, 12 Feet (2 Pack) LAN, utp Cat 6, RJ45, Network Cord, Patch, Internet Cable - 12 ft - Black × 1 ₹999

Delivered 21 Sept, 11:05 am

Order #11442 Delivered 28 Jul, 06:13 pm · NETBANKING (success) · to Mumbai 400049 — **₹2,069**


Adult Arch & Logo Soft Style Gameday T-Shirt × 1 ₹2,069

Delivered 28 Jul, 06:13 pm

Order #10776 Delivered 6 Jul, 07:49 pm · COD (success) · to Mumbai 400049 — **₹3,568**


JETech Screen Protector for iPad 2 3 4 (Oldest Models), Tempered Glass Film, 1-Pack × 1 ₹829


Blue Light Blocking Screen Protector Panel for 14 inch Diagonal LED PC Laptop Anti-UV Eye Protection Filter Film - Widescreen Laptop Frame Hanging Type (W 12.6" X... × 1 ₹2,739

Delivered 6 Jul, 07:49 pm

Order #9794 Delivered 1 Jun, 02:57 pm · CARD (success) · to Mumbai 400049 — **₹3,568**


KALILY 12pcs/9pcs/6pcs Headband Bandana - Face Mask Headwear Neck Gaiter Shield Scarf × 1 ₹1,329

 W WSTOO Snorkel Mask with Latest Dry Top Breathing System,Fold 180 Degree Panoramic View Full Face Snorkel Mask Anti-Fog Anti-Leak with Camera Mount,Snorke... × 1 ₹2,239

Delivered 1 Jun, 02:57 pm

Order #9377 Delivered 17 May, 05:10 pm · CARD (success) · to Mumbai 400049 **₹3,148**

 MOSISO Compatible with MacBook Air 13 inch Case 2022 2021 2020 2019 2018 Release A2337 M1 A2179 A1932 Retina Display with Touch ID, Plastic Hard Shell Case ... × 1 ₹1,489

 KECC Compatible with MacBook Air 13 inch Case 2020 2019 2018 Release A1932 Retina Display + Touch ID Protective Plastic Hard Shell + Keyboard Cover (Black Grey ... × 1 ₹1,659

Delivered 17 May, 05:10 pm

Order #9293 Delivered 14 May, 11:02 pm · UPI (success) · to Mumbai 400049 **₹5,498**

 Wrangler Authentics Men's Classic 5-Pocket Relaxed Fit Flex Jean × 2 ₹5,498

Delivered 14 May, 11:02 pm

Order #8914 Delivered 1 May, 12:20 pm · UPI (success) · to Mumbai 400049 **₹55,356**

 Exercise Bike Trainer - Indoor Bicycle Training Stand With Quiet 5-Level Magnetic Resistance and Front Wheel Riser Block By Bike Lane × 1 ₹5,149

 Gyroor Warrior 8.5 inch All Terrain Off Road Hoverboard with Bluetooth Speakers and LED Lights, UL2272 Certified Self Balancing Scooter × 2 ₹49,798

 Bell Standard and Self Sealing Bike Tubes × 1 ₹409

Delivered 1 May, 12:20 pm

Order #7189 Delivered 28 Feb, 09:02 pm · UPI (success) · to Mumbai 400049 **₹10,788**

 EBL 12 Bay LCD Universal Battery Charger for Rechargeable AA AAA 9V NIMH NICD Batteries × 1 ₹2,489

 DBPOWER 16.9" Portable DVD Player with 14.1" HD Swivel Large Screen, Support DVD/USB/SD Card and Multiple Disc Formats, 6 Hrs 5000mAH Rechargeable Battery, ... × 1 ₹8,299

Delivered 28 Feb, 09:02 pm

Order #6958 Delivered 18 Feb, 07:15 pm · NETBANKING (success) · to Mumbai 400049 **₹5,558**

 VEWOSTAR Garden Solar Wind Spinner - Outdoor Metal Large Wind Sculpture Yard Art Multi-Color LED Lighting Wind Spinner for Garden Patio Lawn × 1 ₹4,069

 YURRUON Metal Craft Ant Yard Decor 4Pcs Ant Metal Sculpture Garden Ant Decoration Hanging Wall Garden Lawn Decoration Indoor and Outdoor Colorful and Lovin... × 1 ₹1,489

Delivered 18 Feb, 07:15 pm

Order #5836 Delivered 31 Dec, 02:31 pm · CARD (success) · to Mumbai 400049 **₹13,608**

 Ringside Pro Style IMF Tech Boxing Training Sparring Gloves × 1 ₹11,289

 Sanabul Essential 7 oz MMA Hybrid Sparring Gloves × 1 ₹2,319

Delivered 31 Dec, 02:31 pm

Order #5706 Delivered 27 Dec, 10:02 am · UPI (success) · to Mumbai 400049 **₹1,748**

 Amazon Basics ABS USB-A to Lightning Cable, MFi Certified Charger Cord for Apple iPhone 14 13 12 11 X Xs Pro, Pro Max, Plus, iPad, 10,000 Bend Lifespan - White, 3-Ft × 1 ₹839

 USB C to Lightning Cable [Apple MFi Certified] 4Pack 3FT iPhone Fast Charger Cable Compatible iPhone 14/14 Pro/14Pro Max/13/13 Pro/12/12 Pro/11/11 Pro/XR/XS/... × 1 ₹909

Delivered 27 Dec, 10:02 am

Order #5210 Delivered 12 Dec, 07:49 am · UPI (success) · to Mumbai 400049 **₹2,259**

 Miesherk iPad Pro 12.9 Case 2022/2021: Military Grade Heavy Duty Shockproof Cover for iPad Pro 12.9 Inch 6th/5th Generation- Pencil Holder - Rotating Stand - Hand... × 1 ₹2,259

Delivered 12 Dec, 07:49 am

Order #5151 Delivered 10 Dec, 06:16 pm · NETBANKING (success) · to Mumbai 400049 **₹1,449**

 Scosche MAGDM2 MagicMount Magnetic Car Phone Holder Mount - 360 Degree Adjustable Head, Universal with All Devices - Dashboard Mount × 1 ₹1,449

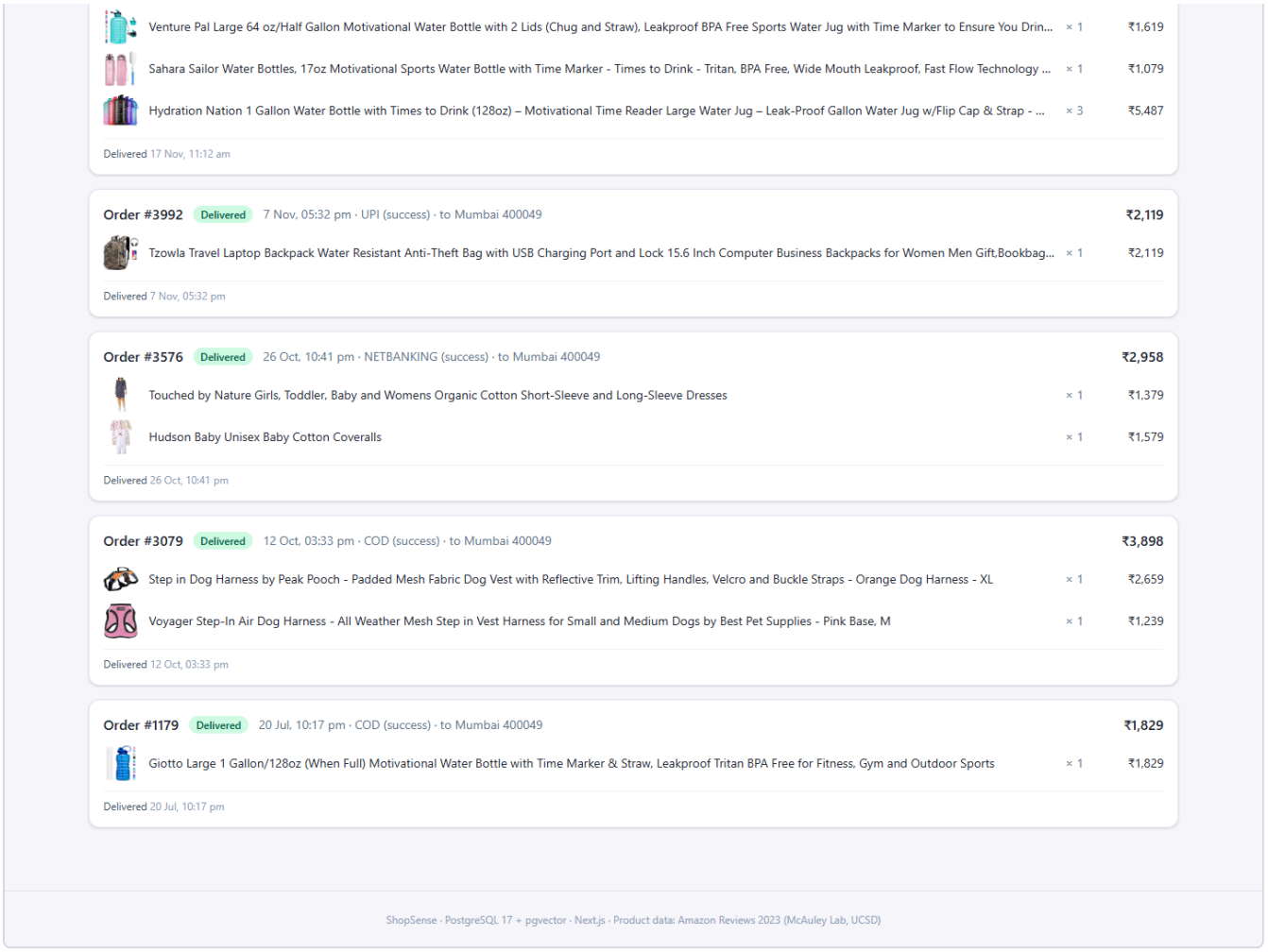
Delivered 10 Dec, 06:16 pm

Order #4638 Delivered 25 Nov, 01:44 pm · UPI (success) · to Mumbai 400049 **₹2,899**

 Playz Edible Poop! Food Science STEM Chemistry Kit - DIY Make Your Own Chocolates and Candy Experiments for Boy, Girls, Teenagers, & Kids Ages 8, 9, 10, 11, 12, 13... × 1 ₹2,899

Delivered 25 Nov, 01:44 pm

Order #4360 Delivered 17 Nov, 11:12 am · UPI (success) · to Mumbai 400049 **₹8,185**



Order history with status timeline from the audit trigger.

ShopSense | Try "gift for a dad who loves gardening" or "quiet keyboard for night typing" | Hybrid | Search | Dashboard | Products | Ayaan Bose · Seller | Sign out

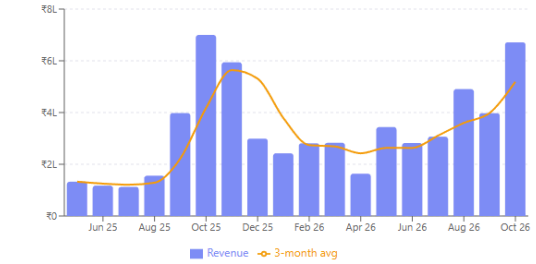
Seller dashboard

Every number here comes from the same views the admin uses. PostgreSQL Row-Level Security limits them to your own products and order lines. The queries contain no `WHERE seller_id = …`.

REVENUE (30 DAYS)	ORDERS (30 DAYS)	UNITS SOLD (30 DAYS)	LIFETIME REVENUE	ACTIVE PRODUCTS
₹7,18,569	151	201	₹59,71,134	128
▲ 93.2% vs previous 30 days				7 low-stock alerts

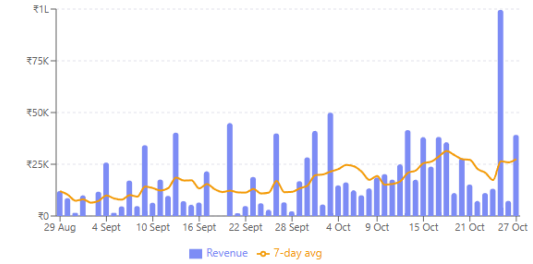
Monthly revenue

Bars: revenue · Line: 3-month moving average (window function)



Daily revenue, last 60 days

Gaps filled with generate_series · Line: 7-day moving average



Month-over-month growth (LAG)

MONTH	REVENUE	ORDERS	MOM
Oct 26	₹6,71,210	142	+69%
Sept 26	₹3,97,226	89	-19%
Aug 26	₹4,90,628	79	+59.9%
Jul 26	₹3,06,774	71	+8.7%
Jun 26	₹2,82,265	61	-17.9%
May 26	₹3,43,613	54	+110.2%
Apr 26	₹1,63,484	35	-42.2%
Mar 26	₹2,82,881	45	+0.7%

Top products, last 90 days

#	PRODUCT	REVENUE	SHARE	RANK IN CATEGORY
1	Sonos Two Room Set One SL -… Portable Audio & Video	₹1,56,845	10%	#1
2	Acer Aspire 3 A314-23P-R3QA… Computers & Accessories	₹1,46,076	9.3%	#1
3	Amazon eero mesh WiFi system … General	₹98,763	6.3%	#1
4	Blink Whole Home Bundle	Vide… General	₹91,295	5.8%	#2
5	All-new Fire HD 10 tablet, 10.1",… General	₹82,134	5.2%	#3
6	NETGEAR C3000-100NAS N300… Computers & Accessories	₹69,414	4.4%	#2
7	Meta Portal - Smart Video Callin… Computers & Accessories	₹64,254	4.1%	#3
8	Amazon Fire HD 10 Tablet Cover… Computers & Accessories	₹46,466	3%	#4
9	Motorola MG7700 Modem WiFi… Computers & Accessories	₹43,557	2.8%	#5
10	StarTech.com 10ft (3m) Comput… Home Audio	₹33,894	2.2%	#1

Low-stock alerts

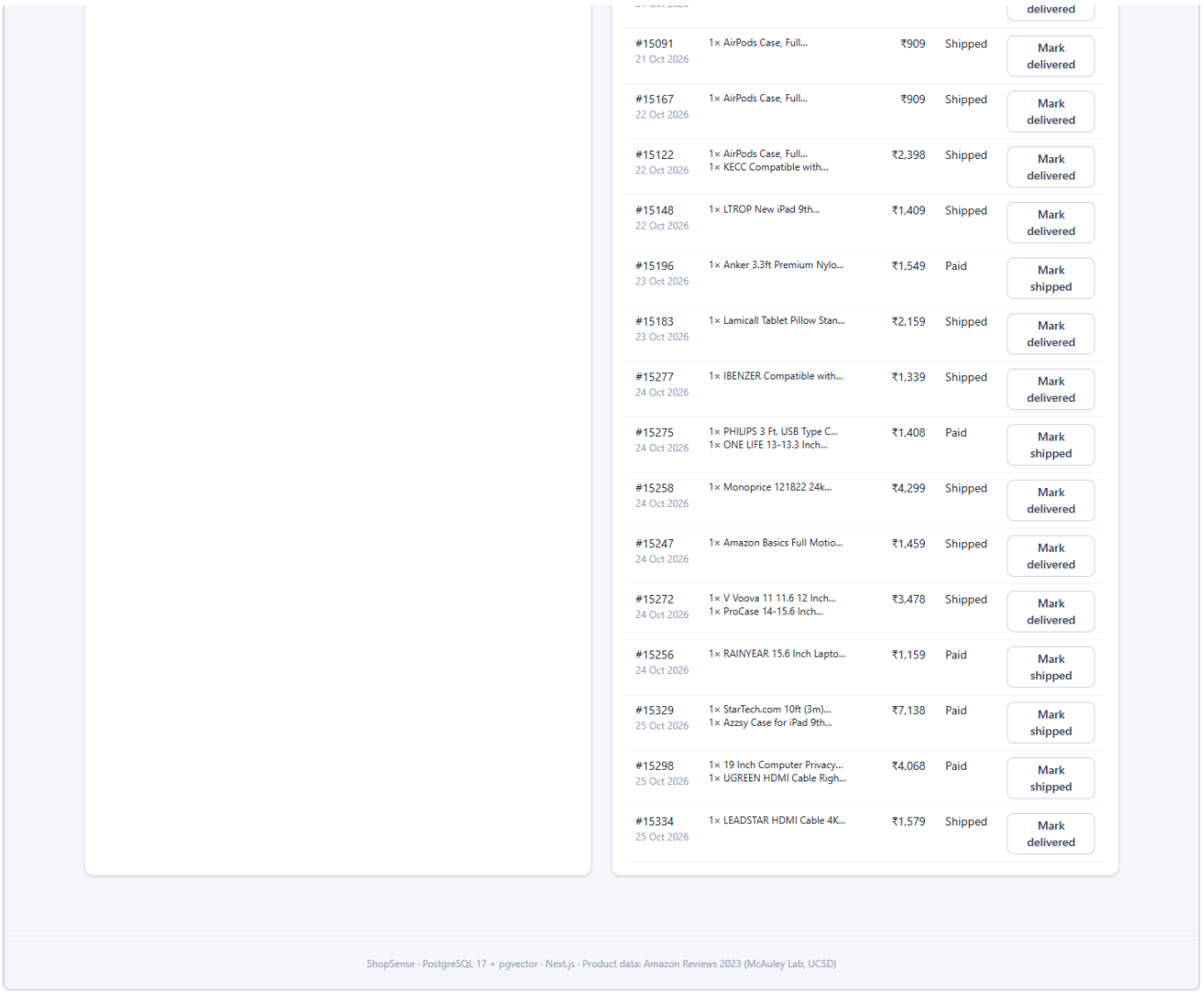
Raised by trigger `trg_products_stock_alert` when stock falls below your threshold; resolved automatically on restock.

PRODUCT	STOCK	THRESHOLD	SINCE		
Fintie Case for iPad Air (3rd Gen) 10.5…	0	10	27 Oct 2026	50	Restock
8K HDMI Cable 15 FT, Capshi Ultra…	1	5	27 Oct 2026	50	Restock
LOVEVOOK Geometric…	1	15	27 Oct 2026	50	Restock
Remote Case for Apple Tv 4th 5th…	1	20	27 Oct 2026	50	Restock
POY Replacement Bands Compatibl…	5	10	27 Oct 2026	50	Restock
Apple Magic Mouse: Wireless,…	6	15	27 Oct 2026	50	Restock
SEKC 120GB SSD SATA III 6 Gb/s, U…	7	15	27 Oct 2026	50	Restock

Orders to fulfil

Only your lines in each order are visible. Illegal status jumps are rejected by trigger.

ORDER	ITEMS	YOUR AMOUNT	STATUS	
#14858 17 Oct 2026	1× Arae Electronic Organizer,… 1× Shintop Cord Organizer, 6…	₹2,078	Shipped	Mark delivered
#14900 18 Oct 2026	1× Sonos Two Room Set One…	₹31,369	Shipped	Mark delivered
#14908 18 Oct 2026	1× Monoprice 121822 24k…	₹4,299	Shipped	Mark delivered
#14942 19 Oct 2026	1× LENRUE Portable Wireless…	₹1,489	Shipped	Mark delivered
#14999 19 Oct 2026	1× Fintie Case for iPad Air (3r…	₹1,159	Shipped	Mark delivered
#14984 19 Oct 2026	2× Scosche MAGDM2…	₹2,898	Shipped	Mark delivered
#15010 20 Oct 2026	1× Blink Whole Home Bundle…	₹18,259	Shipped	Mark delivered
#15041 20 Oct 2026	2× Cablecreation RCA Audio…	₹1,618	Shipped	Mark delivered
#15101 21 Oct 2026	1× Shintop Cord Organizer, 6…	₹499	Shipped	Mark delivered
#15097 21 Oct 2026	1× Monoprice 121822 24k…	₹4,299	Shipped	Mark



Seller dashboard (all numbers restricted by RLS): revenue trend, LAG growth, ranks, low-stock alerts, fulfilment.

ShopSense | Try "gift for a dad who loves gardening" or "quiet keyboard for night typing" | Hybrid | Search | Dashboard | Products | Ayaan Bose · Seller | Sign out

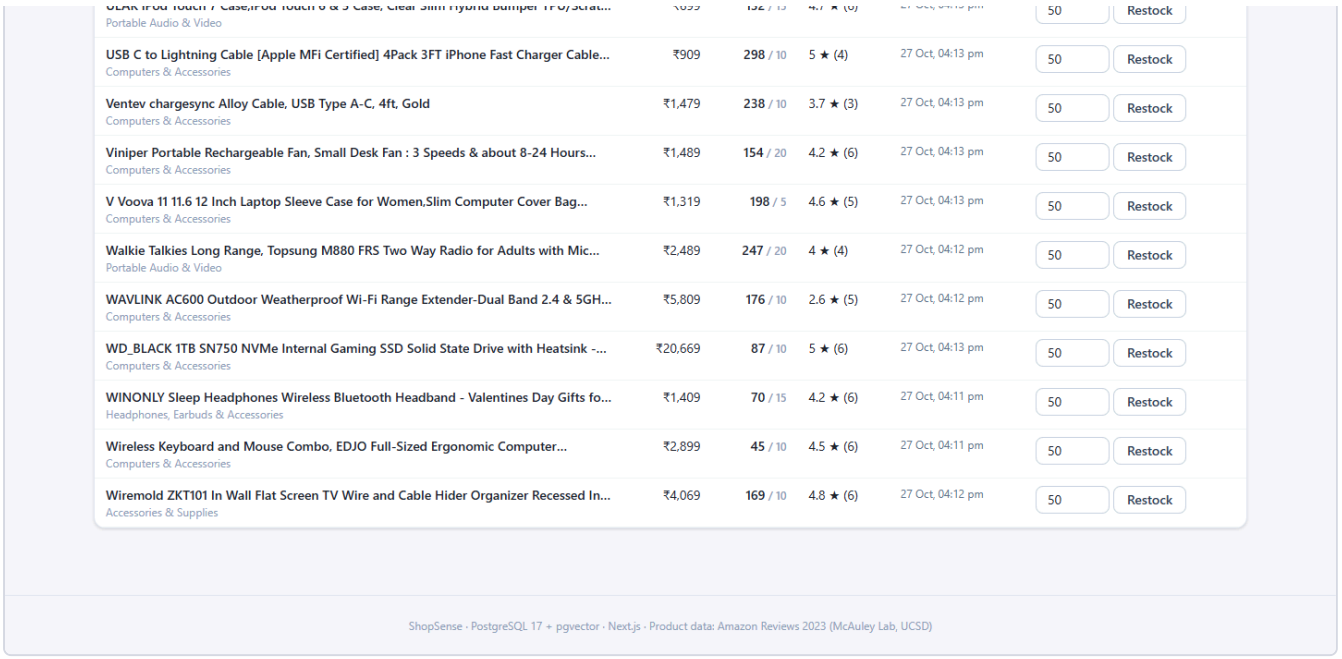
Your products

Filter by title | Filter | + New product

PRODUCT	PRICE	STOCK	RATING	EMBEDDING	RESTOCK
8K HDMI Cable 15 FT, Capshi Ultra 48Gbps High Speed 15 FT HDMI Cable, Long HDM... Television & Video	₹749	1 / 5	4.8 ★ (4)	27 Oct, 04:12 pm	50 Restock
Apple Magic Mouse: Wireless, Bluetooth, Rechargeable. Works with Mac or iPad;... General	₹6,559	6 / 15	3.5 ★ (6)	27 Oct, 04:13 pm	50 Restock
Fintie Case for iPad Air (3rd Gen) 10.5" 2019 / iPad Pro 10.5" 2017 - [Corner Protection... Computers & Accessories	₹1,989	0 / 10	1 ★ (3)	27 Oct, 04:13 pm	50 Restock
LOVEVOOK Geometric Luminous Purses and Handbags for Women Holographic... Computers & Accessories	₹3,149	1 / 15	4.3 ★ (6)	27 Oct, 04:13 pm	50 Restock
POY Replacement Bands Compatible for Fitbit Charge 2, Adjustable Breathable... Wearable Technology	₹609	5 / 10	4 ★ (3)	27 Oct, 04:13 pm	50 Restock
Remote Case for Apple Tv 4th 5th Generation, Akwox Apple TV Siri Remote Cover Ca... Television & Video	₹499	1 / 20	4 ★ (6)	27 Oct, 04:11 pm	50 Restock
SEKC 120GB SSD SATA III 6 Gb/s, Up to 550/530 MB/s Read/Write Speed, Internal 2.5... Computers & Accessories	₹1,329	7 / 15	4.2 ★ (5)	27 Oct, 04:12 pm	50 Restock
19 Inch Computer Privacy Screen Filter for 5:4 Square Computer Monitor - Anti-Glare... Computers & Accessories	₹3,319	115 / 5	3 ★ (5)	27 Oct, 04:13 pm	50 Restock
2 Pack iPhone Earbuds [Apple MFi Certified] Wired in-Ear Stereo Noise Canceling... Headphones, Earbuds & Accessories	₹829	215 / 5	3.5 ★ (6)	27 Oct, 04:12 pm	50 Restock
Acer Aspire 3 A314-23P-R3QA Slim Laptop | 14.0" Full HD IPS Display | AMD Ryzen 5... Computers & Accessories	₹36,519	50 / 10	4.5 ★ (6)	27 Oct, 04:13 pm	50 Restock
AirPods Case, Full Protective Silicone AirPods Accessories Cover Compatible with... Headphones, Earbuds & Accessories	₹909	280 / 20	5 ★ (6)	27 Oct, 04:12 pm	50 Restock
All-new Fire HD 10 tablet, 10.1", 1080p Full HD, 32 GB, latest model (2021 release),... General	₹13,689	199 / 20	4.8 ★ (6)	27 Oct, 04:13 pm	50 Restock
Altec Lansing Mini H2O - Waterproof Bluetooth Speaker, IP67 Certified & Floats in... Portable Audio & Video	₹2,099	289 / 10	5 ★ (6)	27 Oct, 04:13 pm	50 Restock
Amazon Basics 4.7 GB 16x DVD-R (100-Pack Spindle) Computers & Accessories	₹1,629	180 / 10	4.5 ★ (4)	27 Oct, 04:13 pm	50 Restock
Amazon Basics ABS USB-A to Lightning Cable Cord, MFi Certified Charger for Apple... Computers & Accessories	₹789	226 / 20	5 ★ (6)	27 Oct, 04:13 pm	50 Restock
Amazon Basics ABS USB-A to Lightning Cable, MFi Certified Charger Cord for Apple... Computers & Accessories	₹839	134 / 20	4 ★ (6)	27 Oct, 04:13 pm	50 Restock
Amazon Basics Circular Polarizer Camera Lens Filter - 67 mm Camera & Photo	₹1,119	278 / 10	2.8 ★ (6)	27 Oct, 04:13 pm	50 Restock
Amazon Basics Double Nylon Braided USB A Cable with Lightning Connector, Premiu... Computers & Accessories	₹2,289	247 / 5	4.5 ★ (4)	27 Oct, 04:13 pm	50 Restock
Amazon Basics Full Motion Articulating TV Wall Mount Bracket with Swivel and Tilt... Television & Video	₹1,459	294 / 10	4.2 ★ (6)	27 Oct, 04:12 pm	50 Restock
Amazon Basics Lightweight Adjustable Mini Tripod Stand - Pack of 4, 5 - 7.5 Inches,... Camera & Photo	₹2,089	202 / 20	3.5 ★ (6)	27 Oct, 04:12 pm	50 Restock
Amazon Basics USB 2.0 Extension Cable - A-Male to A-Female Adapter Cord - 3.3 Fe... Home Audio	₹2,149	209 / 20	5 ★ (6)	27 Oct, 04:12 pm	50 Restock
Amazon eero mesh WiFi system – router replacement for whole-home coverage (3-... General	₹14,109	97 / 10	4.5 ★ (6)	27 Oct, 04:12 pm	50 Restock
Amazon Fire HD 10 Tablet Cover (Only compatible with 11th generation tablet, 2021... Computers & Accessories	₹3,319	61 / 10	3 ★ (6)	27 Oct, 04:12 pm	50 Restock
Anker 3.3ft Premium Nylon Lightning Cable [2-Pack], Apple MFi Certified for iPhone... Computers & Accessories	₹1,549	294 / 15	4 ★ (6)	27 Oct, 04:12 pm	50 Restock
Arae Electronic Organizer, Travel Cable Organizer, Portable Waterproof Pouch, All-in-... Accessories & Supplies	₹1,579	210 / 20	3 ★ (3)	27 Oct, 04:12 pm	50 Restock
ASURION 3 Year Personal Care Protection Plan ($70 - $79.99) Electronics Warranties	₹659	42 / 10	2.7 ★ (3)	27 Oct, 04:12 pm	50 Restock
AURUM CABLES High Speed HDMI Extension Cable with Ethernet - 4K HDMI Extend... Television & Video	₹909	209 / 20	5 ★ (4)	27 Oct, 04:11 pm	50 Restock
Azzsy Case for iPad 9th Generation/iPad 8th Generation/iPad 7th Generation (10.2... Computers & Accessories	₹1,489	92 / 10	3.2 ★ (5)	27 Oct, 04:12 pm	50 Restock
Battery Organizer Storage Case Holder with Tester BT-168, Batteries Variety Pack Bag,... Household Batteries, Chargers & Accessories	₹2,239	268 / 10	4.8 ★ (6)	27 Oct, 04:13 pm	50 Restock
Best Choice Products 119in Projector Screen 1:1 Indoor Manual Pull Down HD... Television & Video	₹1,019	42 / 10	5 ★ (3)	27 Oct, 04:12 pm	50 Restock
BFRIENDit Wired USB Keyboard with Cover, Comfortable Quiet Chocolate Keys,... Computers & Accessories	₹2,399	289 / 15	4.5 ★ (4)	27 Oct, 04:13 pm	50 Restock
Blink Whole Home Bundle | Video Doorbell System, Outdoor camera, and Mini came... General	₹18,259	240 / 10	4.3 ★ (6)	27 Oct, 04:13 pm	50 Restock
Cablecreation RCA Audio Cable 3.5mm Male to 2-Male RCA Cable, Y Splitter Design... Television & Video	₹809	85 / 20	4.2 ★ (6)	27 Oct, 04:13 pm	50 Restock
CableCreation USB C Cable 10FT USB A to USB C Cable Braided USB to C Cable Fast... Computers & Accessories	₹1,039	264 / 20	4.7 ★ (6)	27 Oct, 04:12 pm	50 Restock
CaseBot Magnetic Case for iPad Pro 12.9-inch 6th Generation 2022, Multiple Angle... Computers & Accessories	₹2,819	155 / 15	5 ★ (3)	27 Oct, 04:12 pm	50 Restock
Cat7 Ethernet Cable - 20 ft - RJ45 Connector - Cat 7 Double Shielded STP - 10 Gigabi... Computers & Accessories	₹1,069	34 / 10	3.7 ★ (3)	27 Oct, 04:12 pm	50 Restock
Cat 8 Ethernet Cable 3 Ft Internet Network Cord, Higher Speed Than Cat 7 Cable, 26... Computers & Accessories	₹409	33 / 10	5 ★ (3)	27 Oct, 04:13 pm	50 Restock

Cat 8 Ethernet Cable 5 ft Shielded, Indoor&Outdoor, Heavy Duty High Speed Direct... Computers & Accessories	₹749	107 / 10	4.8 ★ (6)	27 Oct, 04:13 pm	50	Restock
Chromebook Case, HESTECH 14-15.4 Neoprene Laptop Sleeve Case with Handle for 1... Computers & Accessories	₹1,329	212 / 5	3.7 ★ (6)	27 Oct, 04:11 pm	50	Restock
Corsair CMK32GX4M2F4000C19 Vengeance LPX 32GB (2x16GB) DDR4 4000 C19 1.35V... Computers & Accessories	₹1,419	177 / 10	4 ★ (6)	27 Oct, 04:12 pm	50	Restock
CTCstore 10-in-1 USB 2.0 High Speed Memory Card Reader Writer for SD SDHC Mini... Computers & Accessories	₹1,049	104 / 5	3.3 ★ (3)	27 Oct, 04:12 pm	50	Restock
DACHEE Big White Rose Patten Waterproof Laptop Shoulder Messenger Bag Case... Computers & Accessories	₹1,409	202 / 5	5 ★ (6)	27 Oct, 04:13 pm	50	Restock
DCT Wireless WiFi Security Camera for Outdoor/Home Battery Powered, 2K... Camera & Photo	₹6,639	277 / 20	4.5 ★ (6)	27 Oct, 04:13 pm	50	Restock
Electriduct Cable Shield Cord Cover - CSX-2 PVC Floor Wire Protector - 59 Inch -... Accessories & Supplies	₹3,439	248 / 5	4.4 ★ (5)	27 Oct, 04:13 pm	50	Restock
ExpertPower XP-669 7.5-Inch Dual Band SMA-F Antenna (Black) Television & Video	₹1,019	57 / 15	2.5 ★ (6)	27 Oct, 04:11 pm	50	Restock
Fellowes I-Spire Series Laptop Lift/Stand, White/Gray (9311201) Computers & Accessories	₹2,429	89 / 20	4 ★ (3)	27 Oct, 04:12 pm	50	Restock
Fintie Case for iPad Air (3rd Gen) 10.5" 2019 / iPad Pro 10.5" 2017 - [SlimShell] Ultra... Computers & Accessories	₹1,159	297 / 10	4.8 ★ (6)	27 Oct, 04:12 pm	50	Restock
Fintie Pencil Holder Case for Apple Pencil (1st and 2nd Generation), Soft Neoprene... Computers & Accessories	₹909	161 / 5	4 ★ (3)	27 Oct, 04:13 pm	50	Restock
Fireproof Battery Organizer Storage Case Waterproof & Explosionproof, Safe Bag Fits... Household Batteries, Chargers & Accessories	₹2,319	88 / 10	5 ★ (3)	27 Oct, 04:13 pm	50	Restock
Flash Drive USB Type C Both 3.2 Tech - 2 in 1 Dual Drive Memory Stick High Speed... Computers & Accessories	₹2,479	120 / 15	4.3 ★ (3)	27 Oct, 04:13 pm	50	Restock
Furinno JAYA Large Entertainment Stand for TV Up to 55 Inch, Blackwood Television & Video	₹6,389	30 / 10	3.8 ★ (6)	27 Oct, 04:12 pm	50	Restock
HOTWEEMS Wireless D-09 Computer Mouse USB Cordless Mice for Laptop, Ergo... Computers & Accessories	₹909	68 / 10	4.2 ★ (6)	27 Oct, 04:13 pm	50	Restock
IBENZER Compatible with 2022 M2 MacBook Pro 13 Inch Case 2021 2020 M1 A2338... Computers & Accessories	₹1,339	138 / 10	3 ★ (4)	27 Oct, 04:13 pm	50	Restock
IBENZER Compatible with MacBook Pro 15 Inch Case 2015 2014 2013 2012 A1398, Har... Computers & Accessories	₹1,459	172 / 10	4.3 ★ (6)	27 Oct, 04:13 pm	50	Restock
IBRA Orange HDMI-22ft-UHD HDMI 2.0(4K@60Hz) Ready-18Gbps-28AWG Braided... Television & Video	₹1,699	278 / 10	5 ★ (4)	27 Oct, 04:13 pm	50	Restock
iCasso Compatible with MacBook Pro 13 Inch Case 2016-2020 Release A2338 M1 A22... Computers & Accessories	₹2,319	222 / 20	5 ★ (3)	27 Oct, 04:11 pm	50	Restock
iClever HS01 Food Grade Kids Headphones with Microphone, 85/94dB Volume Limite... Headphones, Earbuds & Accessories	₹1,579	28 / 5	4.5 ★ (6)	27 Oct, 04:12 pm	50	Restock
INSMY Retro Bluetooth Speaker, 20W Portable Wood Home Audio Super Bass Stereo... Portable Audio & Video	₹5,479	52 / 20	5 ★ (3)	27 Oct, 04:13 pm	50	Restock
JSAUX USB C to HDMI Cable 3.3ft/1M, (4K@60Hz) Type-C to HDMI Cord Compatible... Television & Video	₹1,579	298 / 15	3.7 ★ (3)	27 Oct, 04:13 pm	50	Restock
KECC Compatible with MacBook Air 13 inch Case (2010-2017 Release) A1369/A1466... Computers & Accessories	₹2,489	31 / 15	4.3 ★ (3)	27 Oct, 04:11 pm	50	Restock
KECC Compatible with MacBook Air 13 inch Case 2022 2021 2020 2019 2018 Release... Computers & Accessories	₹1,489	209 / 20	3.7 ★ (6)	27 Oct, 04:12 pm	50	Restock
KECC Compatible with MacBook Pro 15 inch Case Cover 2016-2019 Release A1990... Computers & Accessories	₹1,419	206 / 20	4 ★ (6)	27 Oct, 04:12 pm	50	Restock
Kinmac Black Color 360 Degree Protective Canvas Vertical Waterproof Laptop Sleeve... Computers & Accessories	₹1,239	178 / 10	4 ★ (6)	27 Oct, 04:13 pm	50	Restock
Lamicall Tablet Pillow Stand, Pillow Soft Pad for Lap Tablet Holder Dock for Bed with ... Computers & Accessories	₹2,159	174 / 10	4.7 ★ (6)	27 Oct, 04:13 pm	50	Restock
LDKCOK Cat 7 Internet Cable 10ft, Cat7 Outdoor Ethernet Cable 10 ft, 26AWG Heavy-... Computers & Accessories	₹749	81 / 10	5 ★ (6)	27 Oct, 04:11 pm	50	Restock
LEADSTAR HDMI Cable 4K 60HZ 25 Ft, 18Gbps High Speed HDMI 2.0 Cable HDCP 2.2... Television & Video	₹1,579	132 / 10	3 ★ (3)	27 Oct, 04:12 pm	50	Restock
LENRUE Portable Wireless Bluetooth Speaker with Built-in-Mic,Handsfree Call,AUX... Portable Audio & Video	₹1,489	246 / 15	4.3 ★ (6)	27 Oct, 04:13 pm	50	Restock
Lens Cap Bundle - 4 Snap-on Lens Caps for DSLR Cameras - 4 Lens Cap Keepers -... Camera & Photo	₹659	253 / 15	4.5 ★ (6)	27 Oct, 04:11 pm	50	Restock
Logitech M585 Multi-Device Wireless Mouse – Control and Move Text/Images/Files... Computers & Accessories	₹2,389	162 / 5	4.5 ★ (4)	27 Oct, 04:11 pm	50	Restock
Logitech Ultrathin Touch Mouse T630 for Windows 8 Touch Gestures Computers & Accessories	₹9,959	165 / 10	3.8 ★ (6)	27 Oct, 04:12 pm	50	Restock
LOVEVOOK Laptop Backpack for Women Fashion Business Computer Backpacks Trav... Computers & Accessories	₹3,979	193 / 15	5 ★ (6)	27 Oct, 04:12 pm	50	Restock
LTROP New iPad 9th Generation Case 2021, iPad 8th/7th Generation Case for Kids,... Computers & Accessories	₹1,409	169 / 10	4.5 ★ (6)	27 Oct, 04:12 pm	50	Restock
May Chen Compatible with MacBook Air 13 inch Case 2021 2020 2019 2018 Model: M... Computers & Accessories	₹1,579	122 / 15	4 ★ (3)	27 Oct, 04:12 pm	50	Restock
mCover Orange Hard Shell Case with Free Keyboard Cover for 15-inch A1286... Computers & Accessories	₹1,329	121 / 5	5 ★ (4)	27 Oct, 04:11 pm	50	Restock
Meta Portal - Smart Video Calling for the Home with 10" Touch Screen Display - White Computers & Accessories	₹10,709	274 / 5	4.8 ★ (6)	27 Oct, 04:13 pm	50	Restock
MilISO Headphone Splitter for Computer CTIA 3.5mm TRRS Female to Dual TRS Mal... Headphones, Earbuds & Accessories	₹579	149 / 15	5 ★ (4)	27 Oct, 04:13 pm	50	Restock
Monoprice 121822 24k Gold Plated Speaker Banana Plugs, Closed Screw Type (20... Home Audio	₹4,299	220 / 10	3.8 ★ (6)	27 Oct, 04:13 pm	50	Restock

MOSISO Case Only Compatible with MacBook Pro Retina 13 inch (Models: A1502 &... Computers & Accessories	₹1,119	276 / 5	3.8 ★ (6)	27 Oct, 04:13 pm	50	Restock
Motorola Edge | 2022 | 2-Day Battery | Unlocked | Made for US 8/256GB | 50MP... New Arrivals - Electronics	₹29,049	135 / 20	2.5 ★ (6)	27 Oct, 04:13 pm	50	Restock
Motorola MG7700 Modem WiFi Router Combo with Power Boost | Approved by... Computers & Accessories	₹14,519	140 / 5	2.7 ★ (6)	27 Oct, 04:13 pm	50	Restock
NETGEAR C3000-100NAS N300 (8x4) WiFi DOCSIS 3.0 Cable Modem Router (C3000)... Computers & Accessories	₹11,569	46 / 10	4.5 ★ (6)	27 Oct, 04:13 pm	50	Restock
NMEPLAD Projector Screen 150INCH,Portable Movie Screen for Outdoor Indoor,4K... Television & Video	₹2,399	214 / 5	4 ★ (3)	27 Oct, 04:13 pm	50	Restock
Nutale Key Finder, Bluetooth Tracker Item Locator with Key Chain for Keys Pet Wallet... GPS, Finders & Accessories	₹3,319	279 / 10	5 ★ (3)	27 Oct, 04:13 pm	50	Restock
ONE LIFE 13-13.3 Inch Waterproof Neoprene Water Resistent Laptop Sleeve Case... Computers & Accessories	₹999	56 / 10	4.7 ★ (3)	27 Oct, 04:11 pm	50	Restock
Pelican 1050 Micro Case - for iPhone, GoPro, Camera, and more (OD Green) Camera & Photo	₹3,319	236 / 20	4.7 ★ (3)	27 Oct, 04:13 pm	50	Restock
PHILIPS 3 Ft. USB Type C Cable, USB-A to USB-C Red Durable Braided Fast Charging... Computers & Accessories	₹409	119 / 5	4.8 ★ (6)	27 Oct, 04:12 pm	50	Restock
ProCase 14-15.6 Inch Laptop Bag Messenger Shoulder Bag Briefcase Sleeve Case for... Computers & Accessories	₹2,159	160 / 15	5 ★ (4)	27 Oct, 04:11 pm	50	Restock
ProCase Galaxy Tab A 8.0 2017 T380 T385 Case (Old Model), Slim Light Smart Cover... Computers & Accessories	₹909	202 / 10	4.8 ★ (4)	27 Oct, 04:13 pm	50	Restock
Puffer Plus Pop-up Flash Diffuser for Most Sony/Konica/Minolta Camera & Photo	₹969	70 / 20	3.7 ★ (3)	27 Oct, 04:13 pm	50	Restock
QGHXO Band for Garmin Vivofit 4, Soft Silicone Replacement Watch Band Strap for... Wearable Technology	₹1,239	247 / 15	4 ★ (4)	27 Oct, 04:11 pm	50	Restock
RAINYEAR 15.6 Inch Laptop Sleeve Protective Case Soft Lining Zipper Cover Padded... Computers & Accessories	₹1,159	202 / 10	4.2 ★ (6)	27 Oct, 04:12 pm	50	Restock
Ring Stick Up Cam Battery HD security camera with custom privacy controls, Simple... General	₹5,809	28 / 20	4.3 ★ (6)	27 Oct, 04:12 pm	50	Restock
SABRENT USB Type C Aluminum Enclosure for M.2 NVMe SSD in RED (EC-NVME-RED) General	₹2,039	213 / 10	4.8 ★ (4)	27 Oct, 04:13 pm	50	Restock
Samsung micro SDXC UHS-1 Card PRO 64GB Class 10 for Galaxy Note 2 II Computers & Accessories	₹999	114 / 15	3.7 ★ (3)	27 Oct, 04:12 pm	50	Restock
SanDisk 8GB Clip Jam MP3 Player, Blue - microSD Card Slot and FM Radio - SDMX26... Portable Audio & Video	₹2,819	262 / 20	3.7 ★ (6)	27 Oct, 04:12 pm	50	Restock
Scosche MAGDM2 MagicMount Magnetic Car Phone Holder Mount - 360 Degree... GPS, Finders & Accessories	₹1,449	33 / 10	3.8 ★ (4)	27 Oct, 04:12 pm	50	Restock
Sennheiser CX 215 Earphones - Blue Headphones, Earbuds & Accessories	₹2,559	300 / 10	4.7 ★ (3)	27 Oct, 04:13 pm	50	Restock
Shintop Cord Organizer, 6 Pack Black Charger Holder Cable Clips Wire Clips for... Accessories & Supplies	₹499	153 / 10	2.8 ★ (6)	27 Oct, 04:12 pm	50	Restock
SilverStone Technology SST-FF121 120mm Ultra Fine Fan Filter with Magnet... Computers & Accessories	₹539	86 / 15	4.5 ★ (6)	27 Oct, 04:11 pm	50	Restock
Sonos Two Room Set One SL - The Powerful Microphone-Free Speaker for Music and... Portable Audio & Video	₹31,369	116 / 5	4 ★ (3)	27 Oct, 04:12 pm	50	Restock
Sony BDP-S590 3D Blu-ray Disc Player with Wi-Fi (Black) (2012 Model) Television & Video	₹14,939	95 / 10	4.5 ★ (6)	27 Oct, 04:13 pm	50	Restock
Sony LCSCSQ Soft Carrying Case for Sony T, W, and N Series Digital Cameras (Black) Camera & Photo	₹2,999	55 / 5	5 ★ (3)	27 Oct, 04:13 pm	50	Restock
SOUL by Ludacris SL100RB Ultra Dynamic On-Ear Headphones (Discontinued by... FREE One-Day Shipping on Select Electronics with your Discover Card	₹2,199	62 / 10	3.3 ★ (3)	27 Oct, 04:12 pm	50	Restock
Speck Products SeeThru Satin Case for MacBook Pro 15-Inch, Grape Purple (SPK-... Computers & Accessories	₹1,599	274 / 5	5 ★ (4)	27 Oct, 04:12 pm	50	Restock
StarTech.com 10ft (3m) Computer Power Cord, NEMA 5-15P to C13, 10A 125V, 18AWG,... Home Audio	₹5,649	83 / 5	5 ★ (4)	27 Oct, 04:12 pm	50	Restock
StarTech.com 2m (6ft) Long Black Apple 8-pin Lightning Connector to USB Cable for... Computers & Accessories	₹1,489	164 / 15	4.3 ★ (6)	27 Oct, 04:13 pm	50	Restock
SteelSeries Siberia v2 Full-Size Gaming Headset (White) Gaming Accessories	₹8,219	29 / 10	4.6 ★ (5)	27 Oct, 04:13 pm	50	Restock
Surge Protector Power Strip with USB, TROND Ultra Thin Flat Plug 10ft Long Extensio... Accessories & Supplies	₹1,829	153 / 5	4 ★ (6)	27 Oct, 04:12 pm	50	Restock
Thankscase Case Compatible with iPad Pro 12.9 2017&2015 Model, Rotating Case... Computers & Accessories	₹2,489	208 / 15	5 ★ (4)	27 Oct, 04:13 pm	50	Restock
Tobfit 4 Pack Bands Compatible with Fitbit Alta/Alta HR Bands, Soft Sport Silicone... Wearable Technology	₹829	226 / 20	4.8 ★ (6)	27 Oct, 04:13 pm	50	Restock
TOYOUTHS Compatible with Fitbit Sense/Sense 2 Band/Versa 3/Versa 4 Bands Wome... Wearable Technology	₹829	71 / 10	3.5 ★ (4)	27 Oct, 04:11 pm	50	Restock
TrackR bravo - Silver (Discontinued by Manufacturer) GPS, Finders & Accessories	₹2,419	173 / 10	2.3 ★ (6)	27 Oct, 04:12 pm	50	Restock
Treva 5 Inch Portable Fan, Battery Operated or USB Connected Cooling for Travel,... Computers & Accessories	₹1,489	33 / 15	4.3 ★ (3)	27 Oct, 04:13 pm	50	Restock
TwoL Ultra Slim Hard Shell Case and Keyboard Cover Screen Protector for 2021-2018... Computers & Accessories	₹2,159	121 / 10	4 ★ (4)	27 Oct, 04:12 pm	50	Restock
UCERAMI Power Inverter 150W DC 12V to 110V AC Outlet Car Adapter Converter with... Car & Vehicle Electronics	₹1,509	217 / 5	4.7 ★ (3)	27 Oct, 04:13 pm	50	Restock
UESWILL Compatible with MacBook Pro 13 inch 2016-2022, Model A2338 M2 M1... Computers & Accessories	₹1,159	95 / 10	4.3 ★ (3)	27 Oct, 04:12 pm	50	Restock
UGREEN HDMI Cable Right Angle 4K 90 Degree HDMI Cord High Speed Down Angle... Television & Video	₹749	80 / 20	4.8 ★ (4)	27 Oct, 04:12 pm	50	Restock
ULAK iPod Touch 7 Case iPod Touch 6 & 5 Case, Clear Slim Hybrid Bumper TPU/Scrat...	₹699	132 / 15	4.7 ★ (6)	27 Oct, 04:13 pm		



Seller product management with embedding status.

ShopSense | Try "gift for a dad who loves gardening" or "quiet keyboard for ni | Hybrid | Search | Analytics | Search quality | DB lab | Users | Platform Admin · Admin | Sign out

Platform analytics

Served from materialized views (`mv_daily_sales`, `mv_product_pairs`) through admin-only SECURITY DEFINER functions.

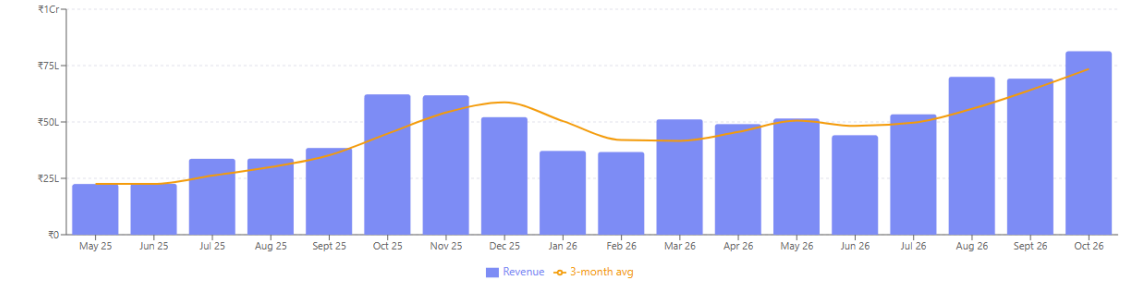
CALL refresh_analytics()

CUSTOMERS	SELLERS	PRODUCTS	ORDERS	REVIEWS	VECTORS STORED	OPEN STOCK ALERTS
800	25	3K	15.4K	14.1K	17.1K pgvector, 384-d	300

Monthly platform revenue

This month so far: ₹81,36,748 · cumulative ₹8,70,64,768

admin_monthly_revenue(): LAG for month-over-month growth, AVG OVER for 3-month moving average, SUM OVER for running total



Nov 25 -0.7% | Dec 25 -15.7% | Jan 26 -28.7% | Feb 26 -1.3% | Mar 26 +39.5% | Apr 26 -4.1% | May 26 +5.1% | Jun 26 -14.4% | Jul 26 +21% | Aug 26 +31.1% | Sept 26 -1.1% | Oct 26 +17.6%

Revenue by category, last 12 months

GROUP BY ROLLUP(category, sub-category): sub-category rows, category subtotals and grand total in one query

CATEGORY	REVENUE	UNITS	SHARE
Beauty and Personal Care	₹48,36,696	2.9K	7.25%
Clothing Shoes and Jewelry	₹74,43,355	3K	11.16%
Electronics	₹1,09,82,156	3.1K	16.46%
Home and Kitchen	₹74,55,279	2.5K	11.18%
Patio Lawn and Garden	₹1,41,59,875	3.3K	21.23%
Pet Supplies	₹53,49,700	2.8K	8.02%
Sports and Outdoors	₹86,22,431	3.1K	12.93%
Toys and Games	₹78,57,799	3.5K	11.78%
All categories	**₹6,67,07,291**	**24.3K**	**100%**

▶ Show sub-category rows (122)

Seller leaderboard, last 6 months

RANK() OVER overall and PARTITION BY main category · share = revenue / SUM() OVER ()

#	STORE	MAIN CATEGORY	REVENUE	SHARE	CAT. RANK
1	**GreenThumb Garden**	Patio Lawn and Garden	₹31,79,739	8.45%	#1
2	**VoltKart Electronics**	Electronics	₹24,97,362	6.64%	#1
3	**Bagicha Supplies**	Patio Lawn and Garden	₹24,03,123	6.39%	#2
4	**Petal & Spade**	Patio Lawn and Garden	₹23,03,412	6.12%	#3
5	**CircuitBay**	Electronics	₹21,20,680	5.64%	#2
6	**Brainy Bricks**	Toys and Games	₹18,20,322	4.84%	#1
7	**Pixel Bazaar**	Electronics	₹17,95,236	4.77%	#3
8	**TrailBlaze Sports**	Sports and Outdoors	₹17,31,256	4.6%	#1
9	**Ghar Essentials**	Home and Kitchen	₹16,95,515	4.51%	#1
10	**FitFuel Gear**	Sports and Outdoors	₹16,72,863	4.45%	#2
11	**Threadline Fashion**	Clothing Shoes and Jewelry	₹16,06,815	4.27%	#1
12	**FunFactory**	Toys and Games	₹14,47,166	3.85%	#2
13	**KitchenKraft**	Home and Kitchen	₹13,38,154	3.56%	#2
14	**SoleStreet**	Clothing Shoes and Jewelry	₹13,22,263	3.52%	#2
15	**Summit Outdoors**	Sports and Outdoors	₹13,06,239	3.47%	#3
16	**HomeNest Living**	Home and Kitchen	₹12,61,861	3.35%	#3
17	**PlayPark Toys**	Toys and Games	₹11,98,044	3.19%	#3
18	**GlowVeda**	Beauty and Personal Care	₹11,36,689	3.02%	#1
19	**Pawsome Supplies**	Pet Supplies	₹11,35,005	3.02%	#1
20	**Kurta & Co**	Clothing Shoes and Jewelry	₹11,17,850	2.97%	#3
21	**TailWag Store**	Pet Supplies	₹10,95,890	2.91%	#2
22	**Whiskers & Paws**	Pet Supplies	₹8,49,294	2.26%	#3
23	**Bloom Beauty**	Beauty and Personal Care	₹8,46,040	2.25%	#2
24	**PureSkin Labs**	Beauty and Personal Care	₹7,32,491	1.95%	#3

Search usage by mode

MODE	SEARCHES	AVG MS	ZERO-RESULT
Lexical	2	141.4	0%
Semantic	2	264.7	0%
Hybrid	15	1221.6	0%

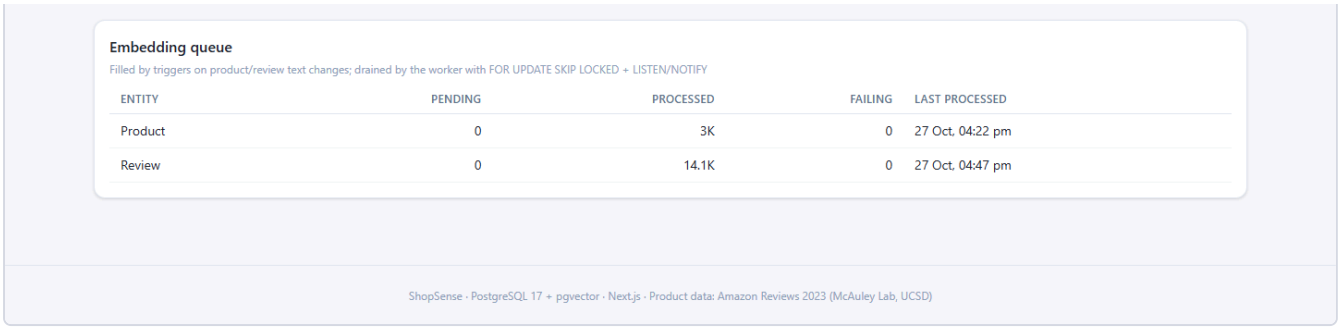
Top queries

something to keep my dog's teeth clean	6
wireless mouse	3
gift for a dad who loves gardening	3
keep coffee hot for hours	3
stainless steel water bottle	2
comfortable shoes for standing all day	2

Zero-result queries

Demand the catalog (or keyword search) is missing

None yet



Admin analytics: ROLLUP, seller leaderboard, search analytics, embedding queue.

Search quality benchmark

40 labelled queries · model Xenova/all-MiniLM-L6-v2 · k = 10 · last run 27 Oct, 08:18 pm. Relevance labels are defined per query as SQL predicates over category and title/description terms (see `eval/queries.json`).

Keyword (full-text)

nDCG@10	0.492
Precision@10	0.472
Recall@10	0.472
MRR	0.813

avg latency 45.14 ms (incl. query embedding)

Semantic (vectors)

nDCG@10	0.660
Precision@10	0.677
Recall@10	0.677
MRR	0.855

avg latency 274.23 ms (incl. query embedding)

Hybrid (RRF)

nDCG@10	0.687
Precision@10	0.710
Recall@10	0.710
MRR	0.880

avg latency 242.46 ms (incl. query embedding)

nDCG@10 by query type

QUERY TYPE	KEYWORD (FULL-TEXT)	SEMANTIC (VECTORS)	HYBRID (RRF)
Natural-language intent queries	0.338	0.575	0.600
Exact / product-name queries	0.852	0.857	0.890

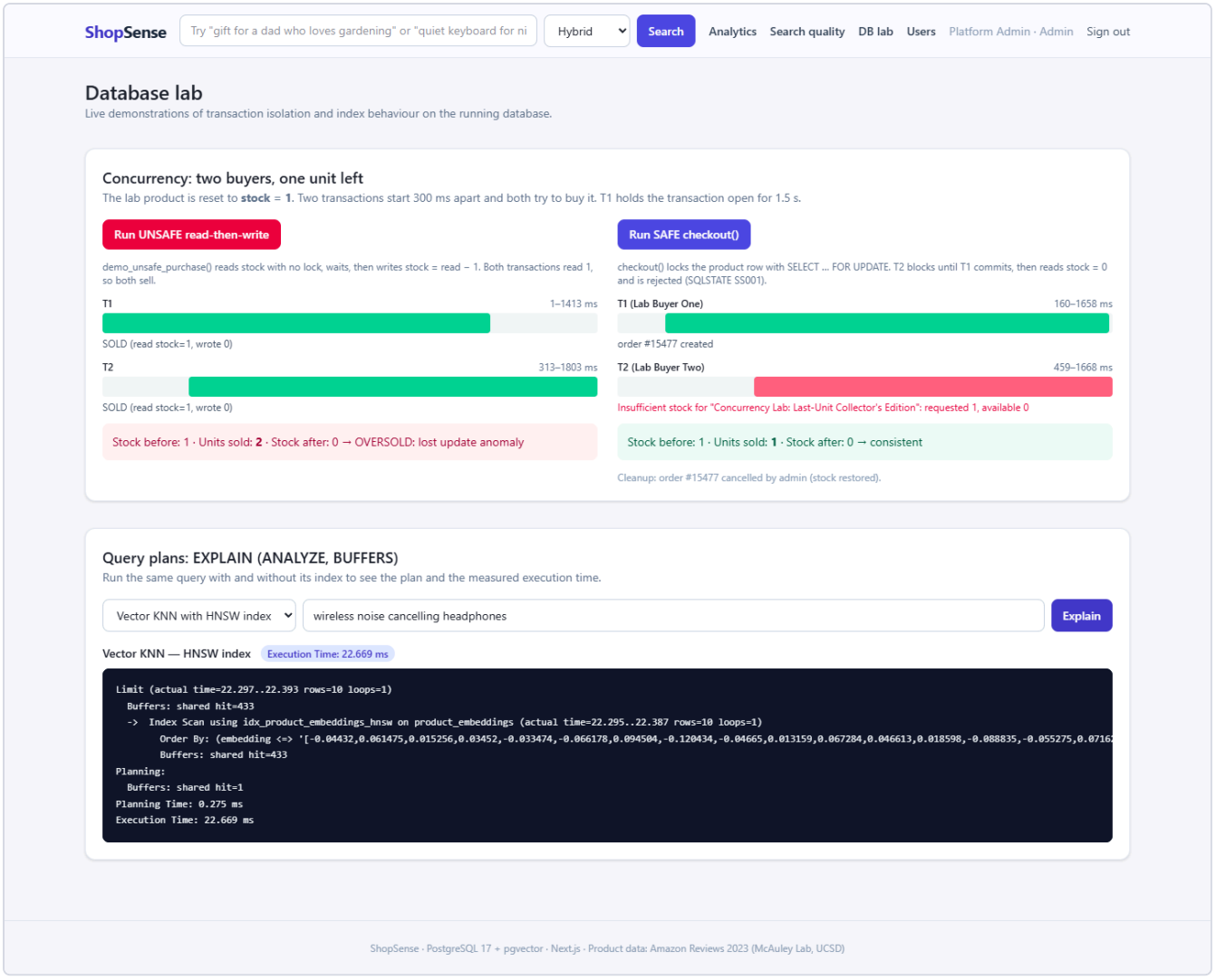
Per-query nDCG@10

QUERY	TYPE	RELEVANT	LEXICAL	SEMANTIC	HYBRID	
something to keep my dog's teeth clean	natural_language	46	0.22	**0.86**	**0.86**	compare
stop my cat from ruining the couch	natural_language	34	**0.40**	0.23	0.23	compare
calm a nervous dog during fireworks	natural_language	39	0.14	0.64	**0.66**	compare
keep my fish tank water clean	natural_language	17	0.00	**0.81**	0.66	compare
gift for a dad who loves gardening	natural_language	22	0.00	**0.10**	0.09	compare
get rid of bugs biting me on the patio	natural_language	30	0.49	0.70	**0.84**	compare
make my backyard glow at night without wiring	natural_language	15	0.00	**0.49**	0.39	compare
attract small birds to my yard	natural_language	31	0.61	0.87	**0.91**	compare
water my lawn automatically	natural_language	85	0.27	**0.90**	0.86	compare
keep coffee hot for hours	natural_language	46	0.64	0.42	**0.71**	compare
soft covers for a good night's sleep	natural_language	106	0.40	0.29	**0.44**	compare
block sunlight so my bedroom stays dark	natural_language	41	0.51	**0.93**	0.93	compare
declutter my small apartment	natural_language	84	0.00	**0.07**	**0.07**	compare
quiet keyboard for typing at night	natural_language	33	0.86	0.87	**0.93**	compare
listen to music without disturbing others	natural_language	43	0.27	0.73	**0.76**	compare
my phone battery dies too fast on trips	natural_language	91	0.54	0.66	**0.71**	compare
take sharp photos of wildlife from far away	natural_language	72	0.06	0.68	**0.71**	compare
workout at home with minimal equipment	natural_language	39	0.29	0.28	**0.33**	compare
sleep outdoors under the stars	natural_language	96	0.07	0.07	**0.09**	compare
protect my sore knees while running	natural_language	56	**0.22**	0.10	0.17	compare
catch more fish this weekend	natural_language	67	0.87	0.80	**0.95**	compare
relief for dry flaky skin in winter	natural_language	158	0.87	0.89	**0.91**	compare
shield my face from the sun	natural_language	17	0.12	**0.68**	0.53	compare
thicker looking hair	natural_language	73	0.43	**0.58**	0.56	compare
toy that teaches kids to code	natural_language	76	0.10	0.39	**0.48**	compare
present for a toddler to develop motor skills	natural_language	133	0.36	0.93	**0.94**	compare
comfortable shoes for standing all day	natural_language	272	0.50	**0.96**	0.77	compare
keep dry in a downpour	natural_language	65	0.22	0.18	**0.29**	compare
dog dental chews	exact	24	**0.87**	0.68	0.87	compare
cat litter	exact	21	**1.00**	**1.00**	**1.00**	compare
solar lights	exact	15	**0.87**	0.80	0.84	compare
pruning shears	exact	12	0.65	0.55	**0.68**	compare
bird feeder	exact	14	0.82	0.80	**0.82**	compare
USB cable	exact	97	**1.00**	**1.00**	**1.00**	compare
HDMI cable	exact	20	0.96	**1.00**	**1.00**	compare
wireless mouse	exact	20	0.89	**0.89**	**0.89**	compare
yoga mat	exact	32	0.59	**0.72**	0.72	compare

memory foam pillow	exact	88	0.65	0.84	0.86	compare
blackout curtains	exact	30	0.92	1.00	1.00	compare
puzzle	exact	28	1.00	1.00	1.00	compare

ShopSense · PostgreSQL 17 + pgvector · Next.js · Product data: Amazon Reviews 2023 (McAuley Lab, UCSD)

Search-quality benchmark results.



Concurrency lab and EXPLAIN ANALYZE viewer.

ShopSense | Try "gift for a dad who loves gardening" or "quiet keyboard for ni | Hybrid | Search | Analytics | Search quality | DB lab | Users | Platform Admin · Admin | Sign out

Users

Name or email | All roles | Search

The app role may update only `users.is_active` (column-level GRANT), only as an admin (RLS policy), and can never read password hashes.

ID	NAME	EMAIL	ROLE	ORDERS	JOINED	LAST LOGIN	STATUS
27	Aarav Sharma	customer@shopsense.dev	Customer	25	19 Mar 2025	27 Oct, 08:41 pm	Active
28	Lab Buyer One	lab.buyer1@shopsense.dev	Customer	1	12 Apr 2025	—	Active
29	Lab Buyer Two	lab.buyer2@shopsense.dev	Customer	0	22 Apr 2025	—	Active
30	Ayaan Verma	ayaan.verma947@example.com	Customer	34	15 Dec 2024	—	Active
31	Neha Bhat	neha.bhat501@example.com	Customer	29	28 Nov 2024	—	Active
32	Harsh Rao	harsh.rao814@example.com	Customer	15	24 Jan 2025	—	Active
33	Aarohi Kapoor	aarohi.kapoor886@example.com	Customer	0	16 Jan 2025	—	Active
34	Arjun Patel	arjun.patel165@example.com	Customer	50	12 Jan 2025	—	Active
35	Karthik Shetty	karthik.shetty665@example.com	Customer	6	24 Mar 2025	—	Active
36	Aditya Verma	aditya.verma846@example.com	Customer	50	27 Feb 2025	—	Active
37	Lakshmi Shetty	lakshmi.shetty389@example.com	Customer	35	14 Mar 2025	—	Active
38	Priya Desai	priya.desai682@example.com	Customer	19	9 Dec 2024	—	Active
39	Krishna Pillai	krishna.pillai473@example.com	Customer	52	27 Oct 2024	—	Active
40	Harsh Bhat	harsh.bhat894@example.com	Customer	39	20 Apr 2025	—	Active
41	Nikhil Reddy	nikhil.reddy309@example.com	Customer	20	9 Dec 2024	—	Active
42	Pooja Das	pooja.das645@example.com	Customer	24	9 Jan 2025	—	Active
43	Aarohi Gupta	aarohi.gupta887@example.com	Customer	0	17 Mar 2025	—	Active
44	Aadhya Gowda	aadhya.gowda241@example.com	Customer	55	17 Oct 2024	—	Active
45	Varun Reddy	varun.reddy569@example.com	Customer	44	21 Dec 2024	—	Active
46	Ishita Malhotra	ishita.malhotra353@example.com	Customer	56	14 Dec 2024	—	Active
47	Kabir Agarwal	kabir.agarwal88@example.com	Customer	14	8 Dec 2024	—	Active
48	Anika Reddy	anika.reddy743@example.com	Customer	3	21 Feb 2025	—	Active
49	Aryan Menon	aryan.menon761@example.com	Customer	5	12 Nov 2024	—	Active
50	Aarav Agarwal	aarav.agarwal462@example.com	Customer	1	5 Feb 2025	—	Active
51	Nikhil Singh	nikhil.singh143@example.com	Customer	36	26 Apr 2025	—	Active
52	Akash Saxena	akash.saxena856@example.com	Customer	25	17 Mar 2025	—	Active
53	Divya Kapoor	divya.kapoor434@example.com	Customer	3	23 Jan 2025	—	Active
54	Aarohi Kulkarni	aarohi.kulkarni251@example.com	Customer	3	2 Feb 2025	—	Active
55	Yash Malhotra	yash.malhotra498@example.com	Customer	11	21 Apr 2025	—	Active
56	Kavya Bose	kavya.bose983@example.com	Customer	83	19 Oct 2024	—	Active
57	Ananya Rao	ananya.rao738@example.com	Customer	1	29 Mar 2025	—	Active
58	Varun Kulkarni	varun.kulkarni293@example.com	Customer	11	15 Dec 2024	—	Active
59	Aadhya Bose	aadhya.bose243@example.com	Customer	21	20 Jan 2025	—	Active
60	Vihaan Khan	vihaan.khan258@example.com	Customer	43	26 Feb 2025	—	Active
61	Arjun Rao	arjun.rao220@example.com	Customer	9	17 Oct 2024	—	Active
62	Myra Sharma	myra.sharma415@example.com	Customer	7	26 Dec 2024	—	Active
63	Aarohi Sharma	aarohi.sharma798@example.com	Customer	37	4 Jan 2025	—	Active
64	Arjun Iyer	arjun.iyer776@example.com	Customer	14	23 Apr 2025	—	Active
65	Shreya Verma	shreya.verma440@example.com	Customer	65	15 Apr 2025	—	Active
66	Nikhil Desai	nikhil.desai601@example.com	Customer	0	6 Nov 2024	—	Active
67	Tejas Verma	tejas.verma44@example.com	Customer	2	20 Jan 2025	—	Active
68	Siddharth Nair	siddharth.nair997@example.com	Customer	1	31 Mar 2025	—	Active
69	Aadhya Gowda	aadhya.gowda299@example.com	Customer	6	11 Nov 2024	—	Active
70	Kabir Menon	kabir.menon711@example.com	Customer	0	30 Jan 2025	—	Active
71	Zara Pillai	zara.pillai203@example.com	Customer	5	8 Dec 2024	—	Active
72	Sai Shetty	sai.shetty849@example.com	Customer	1	25 Oct 2024	—	Active
73	Myra Hegde	myra.hegde747@example.com	Customer	21	7 Jan 2025	—	Active
74	Akash Das	akash.das521@example.com	Customer	5	27 Feb 2025	—	Active
75	Vivaan Das	vivaan.das763@example.com	Customer	33	4 Apr 2025	—	Active
76	Ishaan Reddy	ishaan.reddy174@example.com	Customer	0	27 Jan 2025	—	Active
77	Ishita Saxena	ishita.saxena674@example.com	Customer	14	1 Nov 2024	—	Active
78	Aryan Reddy	aryan.reddy134@example.com	Customer	36	24 Nov 2024	—	Active

79	Aadhya Malhotra	aadhya.malhotra527@example.com	Customer	12	24 Jan 2025	—	Active
80	Pranav Hegde	pranav.hegde665@example.com	Customer	32	29 Apr 2025	—	Active
81	Aarav Fernandes	aarav.fernandes58@example.com	Customer	29	22 Dec 2024	—	Active
82	Sneha Mishra	sneha.mishra200@example.com	Customer	46	18 Dec 2024	—	Active
83	Pranav Menon	pranav.menon564@example.com	Customer	56	8 Mar 2025	—	Active
84	Rahul Kapoor	rahul.kapoor724@example.com	Customer	1	30 Dec 2024	—	Active
85	Vihaan Bhat	vihaan.bhat822@example.com	Customer	35	18 Apr 2025	—	Active
86	Varun Mishra	varun.mishra943@example.com	Customer	52	23 Nov 2024	—	Active
87	Sanya Desai	sanya.desai947@example.com	Customer	33	5 Apr 2025	—	Active
88	Sai Pillai	sai.pillai759@example.com	Customer	0	8 Mar 2025	—	Active
89	Arjun Sharma	arjun.sharma728@example.com	Customer	1	7 Dec 2024	—	Active
90	Sai Rao	sai.rao448@example.com	Customer	3	17 Feb 2025	—	Active
91	Vihaan Nair	vihaan.nair762@example.com	Customer	20	19 Nov 2024	—	Active
92	Pooja Pillai	pooja.pillai595@example.com	Customer	1	22 Dec 2024	—	Active
93	Karthik Reddy	karthik.reddy856@example.com	Customer	26	23 Mar 2025	—	Active
94	Shreya Kapoor	shreya.kapoor582@example.com	Customer	0	3 Feb 2025	—	Active
95	Rahul Verma	rahul.verma26@example.com	Customer	6	25 Nov 2024	—	Active
96	Neha Malhotra	neha.malhotra381@example.com	Customer	5	24 Oct 2024	—	Active
97	Farhan Kapoor	farhan.kapoor368@example.com	Customer	4	18 Dec 2024	—	Active
98	Myra Hegde	myra.hegde423@example.com	Customer	10	25 Nov 2024	—	Active
99	Anika Verma	anika.verma882@example.com	Customer	7	27 Apr 2025	—	Active
100	Sanya Pillai	sanya.pillai883@example.com	Customer	0	21 Dec 2024	—	Active
101	Anika Joshi	anika.joshi801@example.com	Customer	7	28 Nov 2024	—	Active
102	Krishna Agarwal	krishna.agarwal39@example.com	Customer	2	13 Apr 2025	—	Active
103	Aryan Khan	aryan.khan429@example.com	Customer	0	5 Nov 2024	—	Active
104	Nikhil Rao	nikhil.rao971@example.com	Customer	26	8 Feb 2025	—	Active
105	Aryan Mishra	aryan.mishra192@example.com	Customer	50	23 Nov 2024	—	Active
106	Sai Khan	sai.khan944@example.com	Customer	11	20 Oct 2024	—	Active
107	Navya Desai	navya.desai627@example.com	Customer	26	25 Dec 2024	—	Active
108	Sanya Chatterjee	sanya.chatterjee37@example.com	Customer	50	3 Dec 2024	—	Active
109	Vihaan Fernandes	vihaan.fernandes51@example.com	Customer	4	1 May 2025	—	Active
110	Priya Hegde	priya.hegde347@example.com	Customer	22	7 Nov 2024	—	Active
111	Dhruv Reddy	dhruv.reddy697@example.com	Customer	11	28 Dec 2024	—	Active
112	Karthik Reddy	karthik.reddy891@example.com	Customer	2	25 Feb 2025	—	Active
113	Varun Agarwal	varun.agarwal743@example.com	Customer	1	30 Nov 2024	—	Active
114	Vihaan Shetty	vihaan.shetty868@example.com	Customer	10	13 Mar 2025	—	Active
115	Harsh Chatterjee	harsh.chatterjee276@example.com	Customer	36	7 Feb 2025	—	Active
116	Farhan Rao	farhan.rao688@example.com	Customer	10	14 Mar 2025	—	Active
117	Aarohi Gupta	aarohi.gupta689@example.com	Customer	3	30 Jan 2025	—	Active
118	Divya Bhat	divya.bhat367@example.com	Customer	12	2 Mar 2025	—	Active
119	Sanya Khan	sanya.khan385@example.com	Customer	2	15 Jan 2025	—	Active
120	Arjun Iyer	arjun.iyer959@example.com	Customer	9	11 Apr 2025	—	Active
121	Shreya Khan	shreya.khan45@example.com	Customer	0	10 Dec 2024	—	Active
122	Karthik Kulkarni	karthik.kulkarni557@example.com	Customer	4	26 Oct 2024	—	Active
123	Ayaan Gowda	ayaan.gowda798@example.com	Customer	47	16 Jan 2025	—	Active
124	Rahul Reddy	rahul.reddy561@example.com	Customer	0	6 Jan 2025	—	Active
125	Sanya Menon	sanya.menon957@example.com	Customer	31	24 Mar 2025	—	Active
126	Vivaan Reddy	vivaan.reddy901@example.com	Customer	43	26 Jan 2025	—	Active

ShopSense · PostgreSQL 17 + pgvector · Next.js · Product data: Amazon Reviews 2023 (McAuley Lab, UCSD)

Admin user management.

11. Testing

`npm test` runs 22 integration tests against the live database. They connect as the **least-privilege application role**, exactly as the web app does, create their own fixtures and remove them afterwards (the suite was run repeatedly and leaves no data behind).

#	Test	Verifies
1	authenticate	right/wrong password, unknown user, case-insensitive email
2	password storage	bcrypt format; app role cannot `SELECT password_hash` (column grant)
3	register validation	weak password `SS005`; duplicate email
4	domains & CHECKs	bad email, phone, PIN, GSTIN, price ≤ 0, MRP < price, negative stock, rating 6, short review
5	composite FK	a non-seller user cannot become a seller
6	checkout	order, price snapshot, free shipping ≥ ₹499, stock decremented, cart cleared, payment, audit row
7	checkout failures	empty cart `SS002`; over-stock `SS001` via trigger and via stale cart; no partial order left
8	address ownership	cannot ship to someone else's address
9	**concurrency**	two parallel checkouts of the last unit → exactly one succeeds, final stock 0
10, 11	cancel	owner only; stock restored; payment refunded; double-cancel rejected `SS004`
12	order state machine	illegal transitions rejected; history recorded
13	low-stock trigger	alert opens below threshold, updates in place, resolves on restock via procedure
14	RLS (write)	seller cannot update/restock/insert for another seller
15	RLS (read)	customer, both sellers, anonymous see exactly the rows they should; same view gives different revenue
16	RLS (users)	only own row visible; cannot change role; admin-only function rejected
17	review triggers	client-supplied `verified_purchase` overridden; duplicate review; impersonation; aggregates correct
18	embedding queue	text edit enqueues exactly one job; price edit enqueues none
19	association rules	pairs with lift > 1; probabilities within [0, 1]
20	lexical search	finds fixture by rare term; price filter respected
21	hybrid search	ordered by fused score; a product's own vector retrieves it with similarity ≈ 1
22	category helpers	recursive descendants; breadcrumb

Result: 22 passed, 0 failed.

Bugs the tests and the benchmarks caught (all fixed):

1. Trigger functions ran with the caller's rights, so a seller restocking a product (or a customer posting a review) failed with *permission denied* on `stock_alerts` / `embedding_jobs`. Fixed with `SECURITY DEFINER` triggers.
2. The generic embedding-queue trigger read a column that does not exist on one of the two tables.
3. Hybrid fusion weighted noisy keyword matches equally with semantic ones (§9.1).
4. The first fix for (3) made keyword queries ~3× slower, because the planner evaluated the match row by row instead of using the GIN index; restructured with materialized match CTEs (≈100 ms → ≈25 ms).
5. Two test-design errors (an over-strict assumption that a product is its own unique nearest neighbour in a catalog with near-duplicates; reused fixtures making a "not verified" case legitimately verified).

HTTP-level checks (login for each role, role-gated routes, validation messages originating from database constraints) were exercised against the running server.

12. Individual contributions

Member A: transactions, integrity, security & analytics	Member B: search, vectors & recommendations
ER diagram, schema, domains, constraints, indexes	Dataset ingestion; embedding pipeline and queue worker
`checkout`, `cancel_order`, concurrency lab and test	`search_products` (FTS + vector + RRF), search UI, comparison view
Triggers: stock alerts, verified purchase, ratings, state machine, audit	Search-quality benchmark, held-out set, fusion revision
Authentication, roles, Row-Level Security, validation	Frequently-bought-together, similar products, personalisation
Seller and admin dashboards (window functions, ROLLUP)	Product pages, autocomplete, index benchmark, `EXPLAIN` viewer

Commit history in the repository records the individual contributions.

13. Limitations and future work

- Payments, shipping and tax are simulated. Orders, users and carts are synthetic; only product and review text are real (Amazon Reviews 2023, McAuley Lab). Prices are converted from USD at a flat rate; products without a price get a plausible one.
- Relevance labels are pattern-based, not human-annotated; the benchmark is small (58 queries) and the hybrid-vs- semantic difference on unseen queries is within noise. A human-labelled set, a cross-encoder re-ranker, or fine-tuning the embedding model would be the next steps.
- The embedding model is a small general-purpose English model; multilingual queries are out of scope.
- Materialized views are refreshed on demand (`CALL refresh_analytics()`); a scheduled refresh (`pg_cron`) would remove staleness.

14. Conclusion

The project shows a clear connection between database concepts and a working application: the vector database is justified by unstructured product and review text, co-located with relational data so one query combines semantic similarity with business filters; integrity, concurrency and access control are enforced by the database itself (constraints, triggers, row locks, Row-Level Security) rather than by application code; and the claims about search quality are backed by measurement, including a failure we found, diagnosed and fixed, and a held-out set to check the fix. Vector search lifted nDCG@10 from about 0.4–0.5 to 0.66–0.74 over keyword search, and the corrected hybrid design matches or exceeds semantic search while keeping exact-match behaviour.

Appendix A: Data dictionary

addresses

Column	Type	Null	Default / generated	Keys
address_id	bigint	no		PK
user_id	bigint	no		FK → users
label	text	no	'Home'::text	
line1	text	no		
line2	text	yes		
city	text	no		
state	text	no		
pincode	indian_pincode	no		
is_default	boolean	no	false	

CHECK constraints

- `addresses_label_check`: `CHECK ((length(label) <= 30))`
- `addresses_line1_check`: `CHECK ((length(btrim(line1)) >= 3))`

Foreign keys

- `addresses_user_id_fkey`: `FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE`

Indexes

- `CREATE UNIQUE INDEX addresses_pkey ON public.addresses USING btree (address_id)`
- `CREATE INDEX idx_addresses_user ON public.addresses USING btree (user_id)`
- `CREATE UNIQUE INDEX uq_addresses_one_default ON public.addresses USING btree (user_id) WHERE is_default`

Row-Level Security policies

- `addresses_owner` (ALL) USING `((user_id = ( SELECT app_user_id() AS app_user_id)) OR ( SELECT is_admin() AS is_admin))` WITH CHECK `(user_id = ( SELECT app_user_id() AS app_user_id))`

cart_items

Column	Type	Null	Default / generated	Keys
cart_id	bigint	no		PK, FK → carts
product_id	bigint	no		FK → products, PK
quantity	integer	no		

Column	Type	Null	Default / generated	Keys
added_at	timestamp with time zone	no	now()	

CHECK constraints

- `cart_items_quantity_check`: `CHECK (((quantity >= 1) AND (quantity <= 10)))`

Foreign keys

- `cart_items_cart_id_fkey`: `FOREIGN KEY (cart_id) REFERENCES carts(cart_id) ON DELETE CASCADE`
- `cart_items_product_id_fkey`: `FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE`

Indexes

- `CREATE UNIQUE INDEX cart_items_pkey ON public.cart_items USING btree (cart_id, product_id)`
- `CREATE INDEX idx_cart_items_product ON public.cart_items USING btree (product_id)`

Triggers

- `CREATE TRIGGER trg_cart_items_validate BEFORE INSERT OR UPDATE OF quantity ON public.cart_items FOR EACH ROW EXECUTE FUNCTION trg_fn_cart_item_validate()`

Row-Level Security policies

- `cart_items_owner` (ALL) USING `(cart_id = ( SELECT my_cart_id() AS my_cart_id))` WITH CHECK `(cart_id = ( SELECT my_cart_id() AS my_cart_id))`

carts

Column	Type	Null	Default / generated	Keys
cart_id	bigint	no		PK
user_id	bigint	no		UNIQUE, FK → users
updated_at	timestamp with time zone	no	now()	

Foreign keys

- `carts_user_id_fkey`: `FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE`

Indexes

- `CREATE UNIQUE INDEX carts_pkey ON public.carts USING btree (cart_id)`
- `CREATE UNIQUE INDEX carts_user_id_key ON public.carts USING btree (user_id)`

Row-Level Security policies

- `carts_owner` (ALL) USING `(user_id = ( SELECT app_user_id() AS app_user_id))`

categories

Column	Type	Null	Default / generated	Keys
category_id	integer	no		PK
name	text	no		UNIQUE
slug	text	no		UNIQUE
parent_id	integer	yes		UNIQUE, FK → categories

CHECK constraints

- `categories_name_check`: `CHECK ((length(btrim(name)) >= 2))`
- `categories_slug_check`: `CHECK ((slug ~ '^[a-z0-9]+(-[a-z0-9]+)*

Appendix B: Views, functions, procedures and triggers

Generated 2026-10-27.

Views and materialized views

Name	Kind
v_category_tree	VIEW
v_low_stock	VIEW
v_product_cards	VIEW
v_public_reviews	VIEW
v_sales_lines	VIEW
mv_daily_sales	MATERIALIZED VIEW
mv_product_pairs	MATERIALIZED VIEW

Functions and procedures

Name	Kind	Arguments	Returns	Security	Language
refresh_analytics	PROCEDURE			DEFINER	plpgsql
restock_product	PROCEDURE	IN p_product_id bigint, IN p_quantity integer		invoker	plpgsql
admin_category_rollup	FUNCTION	p_since date	TABLE(top_category text, sub_category text, revenue numeric, units bigint, level text, share_pct numeric)	DEFINER	plpgsql
admin_monthly_revenue	FUNCTION		TABLE(month date, revenue numeric, orders bigint, units bigint, mom_growth_pct numeric, moving_avg_3m numeric, cumulative_revenue numeric)	DEFINER	plpgsql
admin_top_sellers	FUNCTION	p_months integer	TABLE(seller_id bigint, store_name text, main_category text, revenue numeric, units bigint, overall_rank bigint, category_rank bigint, revenue_share_pct numeric)	DEFINER	plpgsql

Name	Kind	Arguments	Returns	Security	Language
app_role	FUNCTION		user_role	DEFINER	sql
app_user_id	FUNCTION		bigint	invoker	sql
assert_admin	FUNCTION		void	invoker	plpgsql
assert_password_strength	FUNCTION	p_password text	void	invoker	plpgsql
authenticate	FUNCTION	p_email text, p_password text	TABLE(user_id bigint, full_name text, role user_role)	DEFINER	sql
cancel_order	FUNCTION	p_order_id bigint	void	DEFINER	plpgsql
category_descendants	FUNCTION	p_category_id integer	SETOF integer	invoker	sql
category_path	FUNCTION	p_category_id integer	text	invoker	sql
checkout	FUNCTION	p_address_id bigint, p_method payment_method	bigint	DEFINER	plpgsql
demo_reset_lab_product	FUNCTION		bigint	DEFINER	plpgsql
demo_unsafe_purchase	FUNCTION	p_product_id bigint, p_hold_seconds numeric	text	DEFINER	plpgsql
frequently_bought_together	FUNCTION	p_product_id bigint, p_limit integer	TABLE(product_id bigint, together_count integer, support numeric, confidence numeric, lift numeric)	invoker	sql
is_admin	FUNCTION		boolean	invoker	sql
my_cart_id	FUNCTION		bigint	DEFINER	sql
my_order_ids	FUNCTION		SETOF bigint	DEFINER	sql
my_seller_order_ids	FUNCTION		SETOF bigint	DEFINER	sql
product_embedding_text	FUNCTION	p_product_id bigint	text	invoker	sql
recommend_for_me	FUNCTION	p_limit integer	TABLE(product_id bigint, similarity double precision)	DEFINER	sql
recompute_product_ratings	FUNCTION	p_product_ids bigint[]	void	invoker	sql
refresh_all_taste_profiles	FUNCTION		integer	DEFINER	sql
refresh_taste_profile	FUNCTION	p_user_id bigint	void	DEFINER	sql
register_customer	FUNCTION	p_email text, p_password text, p_full_name text, p_phone text	bigint	DEFINER	plpgsql

Name	Kind	Arguments	Returns	Security	Language
register_seller	FUNCTION	p_email text, p_password text, p_full_name text, p_store_name text, p_gstin text, p_phone text	bigint	DEFINER	plpgsql
review_embedding_text	FUNCTION	p_review_id bigint	text	invoker	sql
search_products	FUNCTION	p_query text, p_embedding vector, p_mode search_mode, p_limit integer, p_category_id integer, p_min_price numeric, p_max_price numeric, p_min_rating numeric, p_in_stock boolean	TABLE(product_id bigint, score double precision, lexical_rank integer, semantic_rank integer, review_rank integer, semantic_similarity double precision, matched_review_id bigint, matched_review_similarity double precision)	invoker	sql
similar_products	FUNCTION	p_product_id bigint, p_limit integer	TABLE(product_id bigint, similarity double precision)	invoker	sql
trending_products	FUNCTION	p_days integer, p_limit integer	TABLE(product_id bigint, units bigint)	DEFINER	sql

Triggers

Table	Trigger	Definition
cart_items	trg_cart_items_validate	`BEFORE INSERT OR UPDATE OF quantity ON public.cart_items FOR EACH ROW EXECUTE FUNCTION trg_fn_cart_item_validate()`
orders	trg_orders_status_guard	`BEFORE UPDATE OF status ON public.orders FOR EACH ROW EXECUTE FUNCTION trg_fn_order_status_guard()`
orders	trg_orders_status_history	`AFTER INSERT OR UPDATE OF status ON public.orders FOR EACH ROW EXECUTE FUNCTION trg_fn_order_status_history()`
orders	trg_orders_touch	`BEFORE UPDATE ON public.orders FOR EACH ROW EXECUTE FUNCTION trg_fn_touch_updated_at()`
products	trg_products_enqueue_embedding_ins	`AFTER INSERT ON public.products FOR EACH ROW EXECUTE FUNCTION trg_fn_enqueue_embedding('product')`

Table	Trigger	Definition
products	trg_products_enqueue_embedding_upd	`AFTER UPDATE OF title, brand, description, features, category_id ON public.products FOR EACH ROW WHEN (((old.title IS DISTINCT FROM new.title) OR (old.brand IS DISTINCT FROM new.brand) OR (old.description IS DISTINCT FROM new.description) OR (old.features IS DISTINCT FROM new.features) OR (old.category_id IS DISTINCT FROM new.category_id))) EXECUTE FUNCTION trg_fn_enqueue_embedding('product')`
products	trg_products_stock_alert	`AFTER INSERT OR UPDATE OF stock_qty, reorder_threshold ON public.products FOR EACH ROW EXECUTE FUNCTION trg_fn_stock_alert()`
products	trg_products_touch	`BEFORE UPDATE ON public.products FOR EACH ROW EXECUTE FUNCTION trg_fn_touch_updated_at()`
reviews	trg_reviews_enqueue_embedding_ins	`AFTER INSERT ON public.reviews FOR EACH ROW EXECUTE FUNCTION trg_fn_enqueue_embedding('review')`
reviews	trg_reviews_enqueue_embedding_upd	`AFTER UPDATE OF title, body ON public.reviews FOR EACH ROW WHEN (((old.title IS DISTINCT FROM new.title) OR (old.body IS DISTINCT FROM new.body))) EXECUTE FUNCTION trg_fn_enqueue_embedding('review')`
reviews	trg_reviews_stats_del	`AFTER DELETE ON public.reviews REFERENCING OLD TABLE AS old_rows FOR EACH STATEMENT EXECUTE FUNCTION trg_fn_review_stats_del()`
reviews	trg_reviews_stats_ins	`AFTER INSERT ON public.reviews REFERENCING NEW TABLE AS new_rows FOR EACH STATEMENT EXECUTE FUNCTION trg_fn_review_stats_ins()`
reviews	trg_reviews_stats_upd	`AFTER UPDATE ON public.reviews REFERENCING OLD TABLE AS old_rows NEW TABLE AS new_rows FOR EACH STATEMENT EXECUTE FUNCTION trg_fn_review_stats_upd()`
reviews	trg_reviews_verify	`BEFORE INSERT ON public.reviews FOR EACH ROW EXECUTE FUNCTION trg_fn_review_verify()`
users	trg_users_normalise	`BEFORE INSERT OR UPDATE OF email, full_name ON public.users FOR EACH ROW EXECUTE FUNCTION trg_fn_users_normalise()`

::text))`

- `chk_category_not_own_parent`: `CHECK ((parent_id IS DISTINCT FROM category_id))`

Foreign keys

- `categories_parent_id_fkey`: `FOREIGN KEY (parent_id) REFERENCES categories(category_id) ON DELETE RESTRICT`

Indexes

- `CREATE UNIQUE INDEX categories_pkey ON public.categories USING btree (category_id)`

- `CREATE UNIQUE INDEX categories_slug_key ON public.categories USING btree (slug)`
- `CREATE INDEX idx_categories_parent ON public.categories USING btree (parent_id)`
- `CREATE UNIQUE INDEX uq_category_name_per_parent ON public.categories USING btree (parent_id, name)`

customer_profiles

Column	Type	Null	Default / generated	Keys
user_id	bigint	no		PK, FK → users
date_of_birth	date	yes		
preferred_category_id	integer	yes		FK → categories
taste_embedding	vector(384)	yes		
taste_updated_at	timestamp with time zone	yes		

CHECK constraints

- `customer_profiles_date_of_birth_check`: `CHECK ((date_of_birth >= '1900-01-01'::date))`

Foreign keys

- `customer_profiles_preferred_category_id_fkey`: `FOREIGN KEY (preferred_category_id) REFERENCES categories(category_id) ON DELETE SET NULL`
- `customer_profiles_user_id_fkey`: `FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE`

Indexes

- `CREATE UNIQUE INDEX customer_profiles_pkey ON public.customer_profiles USING btree (user_id)`

Row-Level Security policies

- `profiles_owner` (ALL) USING `((user_id = ( SELECT app_user_id() AS app_user_id)) OR ( SELECT is_admin() AS is_admin))` WITH CHECK `(user_id = ( SELECT app_user_id() AS app_user_id))`

embedding_jobs

Column	Type	Null	Default / generated	Keys
job_id	bigint	no		PK
entity_type	text	no		
entity_id	bigint	no		
enqueued_at	timestamp with time zone	no	now()	
started_at	timestamp with time zone	yes		
processed_at	timestamp with time zone	yes		
attempts	integer	no	0	
last_error	text	yes		

CHECK constraints

- `chk_jobs_processed_after_enqueue`: `CHECK (((processed_at IS NULL) OR (processed_at >= enqueued_at)))`
- `embedding_jobs_attempts_check`: `CHECK ((attempts >= 0))`
- `embedding_jobs_entity_type_check`: `CHECK ((entity_type = ANY (ARRAY['product'::text, 'review'::text])))`

Indexes

- `CREATE UNIQUE INDEX embedding_jobs_pkey ON public.embedding_jobs USING btree (job_id)`
- `CREATE INDEX idx_embedding_jobs_queue ON public.embedding_jobs USING btree (enqueued_at) WHERE (processed_at IS NULL)`
- `CREATE UNIQUE INDEX uq_embedding_jobs_pending ON public.embedding_jobs USING btree (entity_type, entity_id) WHERE (processed_at IS NULL)`

eval_runs

Column	Type	Null	Default / generated	Keys
run_id	bigint	no		PK
mode	search_mode	no		
model	text	no		
k	integer	no		
query_count	integer	no		
precision_at_k	numeric(5,4)	yes		
recall_at_k	numeric(5,4)	yes		
mrr	numeric(5,4)	yes		
ndcg_at_k	numeric(5,4)	yes		
avg_latency_ms	numeric(10,2)	yes		
details	jsonb	no	'{}'::jsonb	
run_at	timestamp with time zone	no	now()	

CHECK constraints

- `eval_runs_k_check`: `CHECK ((k > 0))`
- `eval_runs_mrr_check`: `CHECK (((mrr >= (0)::numeric) AND (mrr <= (1)::numeric)))`
- `eval_runs_ndcg_at_k_check`: `CHECK (((ndcg_at_k >= (0)::numeric) AND (ndcg_at_k <= (1)::numeric)))`
- `eval_runs_precision_at_k_check`: `CHECK (((precision_at_k >= (0)::numeric) AND (precision_at_k <= (1)::numeric)))`
- `eval_runs_query_count_check`: `CHECK ((query_count > 0))`
- `eval_runs_recall_at_k_check`: `CHECK (((recall_at_k >= (0)::numeric) AND (recall_at_k <= (1)::numeric)))`

Indexes

- `CREATE UNIQUE INDEX eval_runs_pkey ON public.eval_runs USING btree (run_id)`

order_items

Column	Type	Null	Default / generated	Keys
order_id	bigint	no		PK, FK → orders
product_id	bigint	no		PK, FK → products
quantity	integer	no		
unit_price	money_inr	no		
line_total	numeric(12,2)	yes	*generated*	

CHECK constraints

- `order_items_quantity_check`: `CHECK ((quantity > 0))`
- `order_items_unit_price_check`: `CHECK (((unit_price)::numeric > (0)::numeric))`

Foreign keys

- `order_items_order_id_fkey`: `FOREIGN KEY (order_id) REFERENCES orders(order_id) ON DELETE CASCADE`
- `order_items_product_id_fkey`: `FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE RESTRICT`

Indexes

- `CREATE INDEX idx_order_items_product ON public.order_items USING btree (product_id)`
- `CREATE UNIQUE INDEX order_items_pkey ON public.order_items USING btree (order_id, product_id)`

Row-Level Security policies

- `order_items_select` (SELECT) USING `((order_id IN ( SELECT my_order_ids() AS my_order_ids)) OR ( SELECT is_admin() AS is_admin) OR (product_id IN ( SELECT products.product_id FROM products WHERE (products.seller_id = ( SELECT app_user_id() AS app_user_id))))`

order_status_history

Column	Type	Null	Default / generated	Keys
history_id	bigint	no		PK
order_id	bigint	no		FK → orders
old_status	order_status	yes		
new_status	order_status	no		
changed_by	bigint	yes		FK → users
changed_at	timestamp with time zone	no	now()	

Foreign keys

- `order_status_history_changed_by_fkey`: `FOREIGN KEY (changed_by) REFERENCES users(user_id) ON DELETE SET NULL`
- `order_status_history_order_id_fkey`: `FOREIGN KEY (order_id) REFERENCES orders(order_id) ON DELETE CASCADE`

Indexes

- `CREATE INDEX idx_status_hist_order ON public.order_status_history USING btree (order_id, changed_at)`
- `CREATE UNIQUE INDEX order_status_history_pkey ON public.order_status_history USING btree (history_id)`

Row-Level Security policies

- `status_history_select` (SELECT) USING `((order_id IN ( SELECT my_order_ids() AS my_order_ids)) OR ( SELECT is_admin() AS is_admin) OR ((( SELECT app_role() AS app_role) = 'seller'::user_role) AND (order_id IN ( SELECT my_seller_order_ids() AS my_seller_order_ids))))`

orders

Column	Type	Null	Default / generated	Keys
order_id	bigint	no		PK
user_id	bigint	no		FK → users
address_id	bigint	no		FK → addresses
status	order_status	no	'pending'::order_status	
subtotal	money_inr	no		
shipping_fee	money_inr	no	0	
total_amount	money_inr	no		
placed_at	timestamp with time zone	no	now()	
updated_at	timestamp with time zone	no	now()	

CHECK constraints

- `chk_orders_total`: `CHECK (((total_amount)::numeric = ((subtotal)::numeric + (shipping_fee)::numeric)))`

Foreign keys

- `orders_address_id_fkey`: `FOREIGN KEY (address_id) REFERENCES addresses(address_id) ON DELETE RESTRICT`
- `orders_user_id_fkey`: `FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE RESTRICT`

Indexes

- `CREATE INDEX idx_orders_placed ON public.orders USING btree (placed_at)`
- `CREATE INDEX idx_orders_user_placed ON public.orders USING btree (user_id, placed_at DESC)`

- `CREATE UNIQUE INDEX orders_pkey ON public.orders USING btree (order_id)`

Triggers

- `CREATE TRIGGER trg_orders_status_guard BEFORE UPDATE OF status ON public.orders FOR EACH ROW EXECUTE FUNCTION trg_fn_order_status_guard()`
- `CREATE TRIGGER trg_orders_status_history AFTER INSERT OR UPDATE OF status ON public.orders FOR EACH ROW EXECUTE FUNCTION trg_fn_order_status_history()`
- `CREATE TRIGGER trg_orders_touch BEFORE UPDATE ON public.orders FOR EACH ROW EXECUTE FUNCTION trg_fn_touch_updated_at()`

Row-Level Security policies

- `orders_select` (SELECT) USING `((user_id = ( SELECT app_user_id() AS app_user_id)) OR ( SELECT is_admin() AS is_admin) OR ((( SELECT app_role() AS app_role) = 'seller'::user_role) AND (order_id IN ( SELECT my_seller_order_ids() AS my_seller_order_ids))))`
- `orders_update` (UPDATE) USING `(( SELECT is_admin() AS is_admin) OR ((( SELECT app_role() AS app_role) = 'seller'::user_role) AND (order_id IN ( SELECT my_seller_order_ids() AS my_seller_order_ids))))`

payments

Column	Type	Null	Default / generated	Keys
payment_id	bigint	no		PK
order_id	bigint	no		UNIQUE, FK → orders
method	payment_method	no		
amount	money_inr	no		
status	payment_status	no	'pending'::payment_status	
transaction_ref	text	yes		UNIQUE
created_at	timestamp with time zone	no	now()	

Foreign keys

- `payments_order_id_fkey`: `FOREIGN KEY (order_id) REFERENCES orders(order_id) ON DELETE CASCADE`

Indexes

- `CREATE UNIQUE INDEX payments_order_id_key ON public.payments USING btree (order_id)`
- `CREATE UNIQUE INDEX payments_pkey ON public.payments USING btree (payment_id)`
- `CREATE UNIQUE INDEX payments_transaction_ref_key ON public.payments USING btree (transaction_ref)`

Row-Level Security policies

- `payments_select` (SELECT) USING `((order_id IN ( SELECT my_order_ids() AS my_order_ids)) OR ( SELECT is_admin() AS is_admin))`

product_embeddings

Column	Type	Null	Default / generated	Keys
product_id	bigint	no		PK, FK → products
embedding	vector(384)	no		
model	text	no		
source_hash	text	no		
embedded_at	timestamp with time zone	no	now()	

Foreign keys

- `product_embeddings_product_id_fkey`: `FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE`

Indexes

- `CREATE INDEX idx_product_embeddings_hnsw ON public.product_embeddings USING hnsw (embedding vector_cosine_ops) WITH (m='16', ef_construction='64')`
- `CREATE UNIQUE INDEX product_embeddings_pkey ON public.product_embeddings USING btree (product_id)`

products

Column	Type	Null	Default / generated	Keys
product_id	bigint	no		PK
seller_id	bigint	no		FK → sellers
category_id	integer	no		FK → categories
asin	text	yes		UNIQUE
title	text	no		
brand	text	yes		
description	text	no	''::text	
features	text	no	''::text	
price	money_inr	no		
mrp	money_inr	no		
stock_qty	integer	no	0	
reorder_threshold	integer	no	10	
image_url	text	yes		
avg_rating	numeric(2,1)	no	0	
review_count	integer	no	0	
is_active	boolean	no	true	
created_at	timestamp with time zone	no	now()	
updated_at	timestamp with time zone	no	now()	
search_tsv	tsvector	yes	*generated*	

CHECK constraints

- `chk_products_mrp_ge_price`: `CHECK (((mrp)::numeric >= (price)::numeric))`
- `products_avg_rating_check`: `CHECK (((avg_rating >= (0)::numeric) AND (avg_rating <= (5)::numeric)))`
- `products_image_url_check`: `CHECK ((image_url ~ '^https?://'::text))`
- `products_price_check`: `CHECK (((price)::numeric > (0)::numeric))`
- `products_reorder_threshold_check`: `CHECK (((reorder_threshold >= 0) AND (reorder_threshold <= 10000)))`
- `products_review_count_check`: `CHECK ((review_count >= 0))`
- `products_stock_qty_check`: `CHECK ((stock_qty >= 0))`
- `products_title_check`: `CHECK (((length(btrim(title)) >= 3) AND (length(btrim(title)) <= 300)))`

Foreign keys

- `products_category_id_fkey`: `FOREIGN KEY (category_id) REFERENCES categories(category_id) ON DELETE RESTRICT`
- `products_seller_id_fkey`: `FOREIGN KEY (seller_id) REFERENCES sellers(seller_id) ON DELETE RESTRICT`

Indexes

- `CREATE INDEX idx_products_active_price ON public.products USING btree (category_id, price) WHERE is_active`
- `CREATE INDEX idx_products_category ON public.products USING btree (category_id)`
- `CREATE INDEX idx_products_search_tsv ON public.products USING gin (search_tsv)`
- `CREATE INDEX idx_products_seller ON public.products USING btree (seller_id)`
- `CREATE INDEX idx_products_title_trgm ON public.products USING gin (title gin_trgm_ops)`
- `CREATE UNIQUE INDEX products_asin_key ON public.products USING btree (asin)`
- `CREATE UNIQUE INDEX products_pkey ON public.products USING btree (product_id)`

Triggers

- `CREATE TRIGGER trg_products_enqueue_embedding_ins AFTER INSERT ON public.products FOR EACH ROW EXECUTE FUNCTION trg_fn_enqueue_embedding('product')`
- `CREATE TRIGGER trg_products_enqueue_embedding_upd AFTER UPDATE OF title, brand, description, features, category_id ON public.products FOR EACH ROW WHEN (((old.title IS DISTINCT FROM new.title) OR (old.brand IS DISTINCT FROM new.brand) OR (old.description IS DISTINCT FROM new.description) OR (old.features IS DISTINCT FROM new.features) OR (old.category_id IS DISTINCT FROM new.category_id))) EXECUTE FUNCTION trg_fn_enqueue_embedding('product')`
- `CREATE TRIGGER trg_products_stock_alert AFTER INSERT OR UPDATE OF stock_qty, reorder_threshold ON public.products FOR EACH ROW EXECUTE FUNCTION trg_fn_stock_alert()`
- `CREATE TRIGGER trg_products_touch BEFORE UPDATE ON public.products FOR EACH ROW EXECUTE FUNCTION trg_fn_touch_updated_at()`

Row-Level Security policies

- `products_insert` (INSERT) WITH CHECK `(((seller_id = ( SELECT app_user_id() AS app_user_id)) AND (( SELECT app_role() AS app_role) = 'seller'::user_role)) OR ( SELECT is_admin() AS is_admin))`
- `products_select` (SELECT) USING `(is_active OR (seller_id = ( SELECT app_user_id() AS app_user_id)) OR ( SELECT is_admin() AS is_admin))`

- `products_update` (UPDATE) USING `((seller_id = ( SELECT app_user_id() AS app_user_id)) OR ( SELECT is_admin() AS is_admin))` WITH CHECK `((seller_id = ( SELECT app_user_id() AS app_user_id)) OR ( SELECT is_admin() AS is_admin))`

review_embeddings

Column	Type	Null	Default / generated	Keys
review_id	bigint	no		PK, FK → reviews
embedding	vector(384)	no		
model	text	no		
source_hash	text	no		
embedded_at	timestamp with time zone	no	now()	

Foreign keys

- `review_embeddings_review_id_fkey`: `FOREIGN KEY (review_id) REFERENCES reviews(review_id) ON DELETE CASCADE`

Indexes

- `CREATE INDEX idx_review_embeddings_hnsw ON public.review_embeddings USING hnsw (embedding vector_cosine_ops) WITH (m='16', ef_construction='64')`
- `CREATE UNIQUE INDEX review_embeddings_pkey ON public.review_embeddings USING btree (review_id)`

reviews

Column	Type	Null	Default / generated	Keys
review_id	bigint	no		PK
product_id	bigint	no		UNIQUE, FK → products
user_id	bigint	yes		FK → users, UNIQUE
rating	smallint	no		
title	text	yes		
body	text	no		
verified_purchase	boolean	no	false	
helpful_votes	integer	no	0	
created_at	timestamp with time zone	no	now()	

CHECK constraints

- `reviews_body_check`: `CHECK (((length(btrim(body)) >= 10) AND (length(btrim(body)) <= 5000)))`
- `reviews_helpful_votes_check`: `CHECK ((helpful_votes >= 0))`
- `reviews_rating_check`: `CHECK (((rating >= 1) AND (rating <= 5)))`
- `reviews_title_check`: `CHECK ((length(title) <= 200))`

Foreign keys

- `reviews_product_id_fkey`: `FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE`
- `reviews_user_id_fkey`: `FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE SET NULL`

Indexes

- `CREATE INDEX idx_reviews_product ON public.reviews USING btree (product_id, created_at DESC)`
- `CREATE INDEX idx_reviews_user ON public.reviews USING btree (user_id)`
- `CREATE UNIQUE INDEX reviews_pkey ON public.reviews USING btree (review_id)`
- `CREATE UNIQUE INDEX uq_one_review_per_user_product ON public.reviews USING btree (product_id, user_id)`

Triggers

- `CREATE TRIGGER trg_reviews_enqueue_embedding_ins AFTER INSERT ON public.reviews FOR EACH ROW EXECUTE FUNCTION trg_fn_enqueue_embedding('review')`
- `CREATE TRIGGER trg_reviews_enqueue_embedding_upd AFTER UPDATE OF title, body ON public.reviews FOR EACH ROW WHEN (((old.title IS DISTINCT FROM new.title) OR (old.body IS DISTINCT FROM new.body))) EXECUTE FUNCTION trg_fn_enqueue_embedding('review')`
- `CREATE TRIGGER trg_reviews_stats_del AFTER DELETE ON public.reviews REFERENCING OLD TABLE AS old_rows FOR EACH STATEMENT EXECUTE FUNCTION trg_fn_review_stats_del()`
- `CREATE TRIGGER trg_reviews_stats_ins AFTER INSERT ON public.reviews REFERENCING NEW TABLE AS new_rows FOR EACH STATEMENT EXECUTE FUNCTION trg_fn_review_stats_ins()`
- `CREATE TRIGGER trg_reviews_stats_upd AFTER UPDATE ON public.reviews REFERENCING OLD TABLE AS old_rows NEW TABLE AS new_rows FOR EACH STATEMENT EXECUTE FUNCTION trg_fn_review_stats_upd()`
- `CREATE TRIGGER trg_reviews_verify BEFORE INSERT ON public.reviews FOR EACH ROW EXECUTE FUNCTION trg_fn_review_verify()`

Row-Level Security policies

- `reviews_delete` (DELETE) USING `((user_id = ( SELECT app_user_id() AS app_user_id)) OR ( SELECT is_admin() AS is_admin))`
- `reviews_insert` (INSERT) WITH CHECK `((user_id = ( SELECT app_user_id() AS app_user_id)) AND (( SELECT app_role() AS app_role) = 'customer'::user_role))`
- `reviews_select` (SELECT) USING `true`
- `reviews_update` (UPDATE) USING `(user_id = ( SELECT app_user_id() AS app_user_id))` WITH CHECK `(user_id = ( SELECT app_user_id() AS app_user_id))`

search_logs

Column	Type	Null	Default / generated	Keys
log_id	bigint	no		PK
user_id	bigint	yes		FK → users
query	text	no		
mode	search_mode	no		

Column	Type	Null	Default / generated	Keys
filters	jsonb	no	'{}'::jsonb	
result_count	integer	no		
latency_ms	numeric(10,2)	yes		
created_at	timestamp with time zone	no	now()	

CHECK constraints

- `search_logs_latency_ms_check`: `CHECK ((latency_ms >= (0)::numeric))`
- `search_logs_query_check`: `CHECK (((length(query) >= 1) AND (length(query) <= 500)))`
- `search_logs_result_count_check`: `CHECK ((result_count >= 0))`

Foreign keys

- `search_logs_user_id_fkey`: `FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE SET NULL`

Indexes

- `CREATE INDEX idx_search_logs_created ON public.search_logs USING btree (created_at DESC)`
- `CREATE UNIQUE INDEX search_logs_pkey ON public.search_logs USING btree (log_id)`

Row-Level Security policies

- `search_logs_insert` (INSERT) WITH CHECK `((user_id IS NULL) OR (user_id = ( SELECT app_user_id() AS app_user_id)))`
- `search_logs_select` (SELECT) USING `( SELECT is_admin() AS is_admin)`

sellers

Column	Type	Null	Default / generated	Keys
seller_id	bigint	no		PK, FK → users
role	user_role	no	'seller'::user_role	FK → users
store_name	text	no		UNIQUE
gstin	gstin_code	yes		UNIQUE
support_email	email_address	yes		
is_verified	boolean	no	false	
joined_at	timestamp with time zone	no	now()	

CHECK constraints

- `sellers_role_check`: `CHECK ((role = 'seller'::user_role))`
- `sellers_store_name_check`: `CHECK ((length(btrim(store_name)) >= 2))`

Foreign keys

- `fk_sellers_user_is_seller`: `FOREIGN KEY (seller_id, role) REFERENCES users(user_id, role) ON UPDATE RESTRICT ON DELETE CASCADE`

Indexes

- `CREATE UNIQUE INDEX sellers_gstin_key ON public.sellers USING btree (gstin)`
- `CREATE UNIQUE INDEX sellers_pkey ON public.sellers USING btree (seller_id)`
- `CREATE UNIQUE INDEX sellers_store_name_key ON public.sellers USING btree (store_name)`

stock_alerts

Column	Type	Null	Default / generated	Keys
alert_id	bigint	no		PK
product_id	bigint	no		FK → products
stock_qty	integer	no		
threshold	integer	no		
created_at	timestamp with time zone	no	now()	
resolved_at	timestamp with time zone	yes		

CHECK constraints

- `chk_alert_resolved_after_created`: `CHECK (((resolved_at IS NULL) OR (resolved_at >= created_at)))`
- `stock_alerts_stock_qty_check`: `CHECK ((stock_qty >= 0))`
- `stock_alerts_threshold_check`: `CHECK ((threshold >= 0))`

Foreign keys

- `stock_alerts_product_id_fkey`: `FOREIGN KEY (product_id) REFERENCES products(product_id) ON DELETE CASCADE`

Indexes

- `CREATE UNIQUE INDEX stock_alerts_pkey ON public.stock_alerts USING btree (alert_id)`
- `CREATE UNIQUE INDEX uq_stock_alerts_open ON public.stock_alerts USING btree (product_id) WHERE (resolved_at IS NULL)`

Row-Level Security policies

- `stock_alerts_select` (SELECT) USING `(( SELECT is_admin() AS is_admin) OR (product_id IN ( SELECT products.product_id FROM products WHERE (products.seller_id = ( SELECT app_user_id() AS app_user_id))))`

users

Column	Type	Null	Default / generated	Keys
user_id	bigint	no		PK, UNIQUE

Column	Type	Null	Default / generated	Keys
email	email_address	no		UNIQUE
password_hash	text	no		
full_name	text	no		
phone	indian_phone	yes		
role	user_role	no	'customer'::user_role	UNIQUE
is_active	boolean	no	true	
created_at	timestamp with time zone	no	now()	
last_login_at	timestamp with time zone	yes		

CHECK constraints

- `chk_users_hash_is_bcrypt`: `CHECK ((password_hash ~~ '$2%'::text))`
- `users_full_name_check`: `CHECK (((length(btrim(full_name)) >= 2) AND (length(btrim(full_name)) <= 100)))`

Indexes

- `CREATE UNIQUE INDEX uq_users_email ON public.users USING btree (email)`
- `CREATE UNIQUE INDEX uq_users_email_lower ON public.users USING btree (lower((email)::text))`
- `CREATE UNIQUE INDEX uq_users_id_role ON public.users USING btree (user_id, role)`
- `CREATE UNIQUE INDEX users_pkey ON public.users USING btree (user_id)`

Triggers

- `CREATE TRIGGER trg_users_normalise BEFORE INSERT OR UPDATE OF email, full_name ON public.users FOR EACH ROW EXECUTE FUNCTION trg_fn_users_normalise()`

Row-Level Security policies

- `users_admin_update` (UPDATE) USING `( SELECT is_admin() AS is_admin)` WITH CHECK `( SELECT is_admin() AS is_admin)`
- `users_select` (SELECT) USING `((user_id = ( SELECT app_user_id() AS app_user_id)) OR ( SELECT is_admin() AS is_admin))`

Domains (reusable validation)

Domain	Base type	Constraint
cardinal_number	integer	`CHECK ((VALUE >= 0))`
email_address	text	`CHECK ((VALUE ~* '^[a-z0-9._%+-]+@[a-z0-9.-]+.[a-z]{2,}

Appendix B: Views, functions, procedures and triggers

{{SCHEMA_REFERENCE}} ::text)) | `| gstin_code | text |` CHECK ((VALUE ~ '^[0-9]{2}[A-Z]{5}[0-9]{4}[A-Z][1-9A-Z]Z[0-9A-Z]

Appendix B: Views, functions, procedures and triggers

{{SCHEMA_REFERENCE}} ::text)) | `| indian_phone | text |` CHECK ((VALUE ~ '^[6-9][0-9]{9}

Appendix B: Views, functions, procedures and triggers

{{SCHEMA_REFERENCE}} ::text)) | | `indian_pincode | character(6) |` CHECK ((VALUE ~ '^[1-9][0-9]{5}

Appendix B: Views, functions, procedures and triggers

{{SCHEMA_REFERENCE}} ::text)) | | `money_inr | numeric(12,2)` | CHECK ((VALUE >= (0)::numeric)) | | `yes_or_no | character varying(3)` | CHECK (((VALUE)::text = ANY ((ARRAY['YES'::character varying, 'NO'::character varying])::text[])))` |

Enumerated types

Type	Values
order_status	pending, paid, shipped, delivered, cancelled
payment_method	upi, card, netbanking, cod
payment_status	pending, success, failed, refunded
search_mode	lexical, semantic, hybrid
user_role	customer, seller, admin

Appendix B: Views, functions, procedures and triggers

{{SCHEMA_REFERENCE}}